ISERVE+: A PESO LOS BAÑOS SERVICE MANAGEMENT PROCESS-BASED HIRING PLATFORM WITH AI-POWERED UPSKILLING RECOMMENDATION FOR BLUE-COLLAR WORKERS

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Specialized in Service Management Program

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DEDICATION

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ABSTRACT

iServe+, a process-based service management platform designed to optimize hiring and workforce development for the blue-collar around Los Baños. It is a multi-platform system that can suggest work available to workers that matches their skills. Using AI, it simplifies the recruitment process by intelligently matching workers' skills with employers' needs. In addition, iServe+ introduces AI-driven upskilling pathways that provide personalised training recommendations to address skills mismatches and promote continuous learning. The system uses content-based filtering to match job seekers based on their skills. Developed using Laravel and supported by technologies like MySQL, JavaScript, CSS, and Visual Studio Code. To assess usability and functionality, 50 respondents, with 25 workers and 25 clients, participated in an evaluation using the Technology Acceptance Model (TAM) and ISO 25010 evaluation, resulting in a "Good" rating of 4, reflecting the system's dependability, adaptability and performance efficiency. These positive evaluations highlight the system's effectiveness in handling workers and clients and ensuring a reliable user experience. The platform was designed to promote sustainable job creation and workforce development. The study demonstrated how iServe+ had the potential to function as a scalable model for addressing systemic challenges in blue-collar employment while supporting inclusive economic growth.

Keywords: *iServe+, Artificial Intelligence, Job Matching System, Blue Collar Employment, Workforce Development, Content Based Filtering, Technology Acceptance Model, ISO 25010*

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DEFINITION OF TERMS

Artificial Intelligence (AI)	A branch of computer science focused on creating systems capable of performing tasks that typically require human intelligence, such as problem-solving, learning, and decision-making.
Blue Collar	Refers to workers who engage in manual labor, often in industries such as manufacturing, construction, and maintenance, where physical work is a primary component of their job.
Categorization	The process of grouping jobs based on the type of work performed, typically distinguishing between manual labor and skilled trades.
Client	A person who is receiving the benefits, services or service provider.
Education	The systematic transmission of knowledge, skills, and character development, often facilitated through formal instruction in schools, colleges, or vocational training programs.
Job Street	A prominent online job platform that connects employers with job seekers, allowing employers to post job openings and candidates to apply for various positions.
LinkIn	A professional networking platform designed for career and business professionals to connect, share industry insights, and explore job opportunities.
Manual Labor	Physical work performed by individuals, often involving repetitive tasks that require strength, endurance, and physical skill.
PESO (Public Employment Service Office)	A government agency in the Philippines that facilitates job placement services, connects job seekers with employment opportunities, and provides skills development programs.

SDG (Sustainable Development Goals)	A set of 17 global goals established by the United Nations to address pressing social, economic, and environmental challenges, with specific targets to be achieved by 2030.
Skilled Trade	Occupations that require specialized training and technical knowledge, often involving hands-on work in fields such as plumbing, electrical work, and carpentry.
Technical Education and Skills Development Authority (TESDA)	A government agency in the Philippines responsible for managing and supervising technical education and skills development programs, as well as monitoring workforce training initiatives.
Unemployed	An individual who is not currently employed and is actively seeking work but is unable to secure a job.
White Collar	Refers to workers who perform professional, managerial, or administrative tasks in an office environment, typically requiring mental rather than physical effort.
Worker	Refers to a person who is employed to do physical or mental work for wages

CHAPTER 1

INTRODUCTION

Project Context

The majority of the skilled workforce in Los Baños comprises blue-collar workers who usually work in manual labour, technical, or trade jobs. These workers are very practical and specialized in building and construction, maintenance, electrical work, plumbing, agriculture, and manufacturing, among others. However, many skilled blue-collar workers are unable to secure decent work because of limited job-matching services, outdated labour information, and fewer opportunities for continuous training. Many of them end up underemployed or unemployed, which warrants the creation of programs that link them to available job openings and further enhance their competencies.

According to PESO Los Baños, 2,179 of more than 3,000 skilled workers in the Los Baños community are unemployed or do not have permanent jobs. Often, more are unemployed than employed within a year, also due to the lack of sources and information. Another reason is their high qualifications, which make them more likely to remain on the sidelines. That is why our school has collaborated more with the Los Baños community to facilitate their access to employment. This is also a good opportunity for the people of Los Baños to be more productive and not waste their talents and skills.

The Philippines faces significant unemployment challenges, with approximately 4 million workers (12% of the labor force) unemployed and 5

million (17% of those employed) underemployed (Felipe & Lanzona, 2006). This issue persists despite efforts to export labor, which has not effectively reduced unemployment in key occupations (Stahl, 1988). The government has implemented employment and skills development programs through Public Employment Service Offices (PESOs) to address this problem (Banque Asiatique de Développement, 2017). A time-series analysis of unemployment in the Philippines revealed that the labour force rate and population are significant factors affecting unemployment rates (Urrutia et al., 2017). The study also found Granger-causal relationships among unemployment, population, GDP, and GNI. While remittances from overseas workers provide foreign exchange, their impact on economic development is limited due to low investment propensity and diversion through informal channels (Stahl, 1988).

Blue-collar workers play a critical role in national and local economies. Yet, they continue to face challenges related to unstable employment, skills mismatch, and limited access to digital job-finding platforms. A study by Ayudhya & Lewis (2020) highlights that blue-collar workers often experience job insecurity due to unstable labor markets and lack of structured career pathways. Similarly, research by Smith, Wessel, & Jenkins (2022) found that blue-collar workers struggle with underemployment and irregular job opportunities, which are worsened by limited access to formal hiring systems that typically benefit white-collar applicants.

In the Philippine context, the International Labor Organization (ILO) reports that many blue-collar workers lack proper channels to access job opportunities, especially in skilled trades such as construction, repair, and

technical services. This gap is worsened by the absence of digital systems that can efficiently connect workers to employers in real time.

Blue-collar work, long equated with manual labor and skilled trades, is vital to the economy. As Dhir (2025) has observed, these jobs are vital to a range of industries, but are themselves frequently subject to job insecurity and skills deficits. Unlike white-collar employment, which is often characterized by higher educational requirements and more secure career progression, blue-collar work frequently lacks comparable levels of job security and career mobility, making it essential to find solutions to close these gaps.

The fast pace of mobile application development has revolutionized the way services are provided, allowing blue-collar workers to gain access to employment opportunities and training materials more easily (Tracy, 2012). Compared to conventional job search techniques that are time-consuming and have limited scope, mobile applications offer instant access to a wider range of opportunities. This is in contrast to the relatively static nature of white-collar job postings, where networking and official applications tend to rule the hiring scene.

Current developments in recommendation systems have shown promise for enhancing job matching processes. According to Chen and Lee (2024), recommendation systems are able to link employees to suitable opportunities based on their abilities and preferences.

This is compared with conventional recruitment strategies that mostly rely on subjective criteria and social connections, which discriminate against

blue-collar workers. Through the use of data-driven techniques, iServe+ hopes to build a more level playing field job matching

Objectives of the Study

The main objective of the project is to create an integrated service management platform that leverages AI-driven upskilling pathways to enhance the employability and skill development of blue-collar workers while facilitating efficient hiring processes for employers.

1. To design and develop an integrated hiring and job-finding platform that connects employers and workers through job requirements, worker profiles, skills, availability and rates.

2. Design and implement a machine learning-based recommendation engine that suggests upskilling paths for workers based on market trends, feedback and successful profiles.

3. Test and evaluate the acceptability of the application through a software review using ISO 25010 and the Technology Acceptance Model.

Project Purpose

The purpose of this project is to develop a multi-platform job-matching and career-assistance system specifically designed for blue-collar workers in Los Baños. This system is implemented for the Public Employment Service Office (PESO) to enhance local employment opportunities and improve the hiring process for both job seekers and employers. The platform provides an efficient job-matching system that connects job seekers with relevant employment opportunities based on their skills, helps job seekers assess

employers and job listings, and promotes transparency and trust in the local job market. Additionally, the system integrates location-based job search features to help job seekers find high-demand opportunities within Los Baños. Employment agencies and businesses can also utilise the platform to gain data-driven insights for better workforce planning and decision-making.

By streamlining the recruitment process, this platform aims to improve employment rates within Los Baños and assist PESO in providing better job placement services, ultimately contributing to the growth of the local workforce and economy.

The project aligns with the United Nations Sustainable Development Goals (SDGs), addressing multiple objectives, including Goal 1: No Poverty, by providing equitable access to employment opportunities and helping individuals secure stable incomes. Goal 4, Quality Education by emphasizing the value skills, certifications, and training, encouraging continuous learning and professional growth. Goal 8, Decent Work and Economic Growth, by enhancing the quality of job placements and fostering inclusive economic development. Goal 10, Reduced Inequalities by ensuring that workers from diverse backgrounds can access opportunities aligned with their skills and aspirations. Goal 17, Partnerships for the Goals, by collaborating with employers, training institutions, and other stakeholders to strengthen job-matching frameworks globally.

This project benefits individuals by providing fair access to meaningful work, employers by simplifying the recruitment process, and society at large by fostering economic inclusivity and sustainable development.

Blue-Collar Candidates. This research is important to blue-collar job applicants in Los Baños, as it sheds more light on the local labour market, including high-demand jobs and the skills employers require. By identifying current issues such as job mismatches, restricted training access, and insufficient job information, the research aims to help job applicants make better career choices. In addition, the recommendations from this study can help increase access to job opportunities, advance career preparedness, and enhance the overall quality of life for blue-collar workers.

Employers. This research provides useful information on the availability, skills, and requirements of the blue-collar workers in Los Baños. It assists in identifying labour supply gaps, recruitment issues, and potential approaches to working with PESO to enhance the local talent pool. The results can assist employers in improving recruitment practices, customizing on-the-job training programs, and supporting workforce development efforts. Finally, the research helps companies build a more stable, competent, and job-ready workforce to sustain productivity and growth.

PESO (Public Employment Service Office). This project is significant as it seeks to contribute meaningful insights to the Public Employment Service Office (PESO) of Los Baños, which plays a vital role in facilitating local employment opportunities and ensuring workforce alignment within the municipality. The study focuses on the status of blue-collar jobs in Los Baños to identify prevailing trends, challenges, and potential development areas in the local labour market.

Through this research, PESO Los Baños may benefit from:

A clearer understanding of the current supply-and-demand dynamics for blue-collar occupations in the municipality.

Data-driven recommendations that can enhance existing employment facilitation strategies, job matching services, and training initiatives.

The findings of this study may serve as a valuable reference for:

Job seekers can benefit from insights into emerging opportunities and the necessary competencies for blue-collar employment.

Employers and industry partners may use the data to guide recruitment practices and collaboration with PESO and vocational institutions.

Local government officials and policymakers are involved in crafting informed labour policies and supporting sustainable workforce development in the municipality.

By aligning with PESO Los Baños' objectives, this study aims to support the office in its mission to promote inclusive and accessible employment, particularly for blue-collar workers, who are a significant part of the local economy.

Future Researchers. This study may benefit future researchers by providing a model for the community that offers valuable insights into how digital platforms are reshaping the employment landscape for blue-collar workers. It highlights the technology's role in shaping workforce dynamics, including recruitment processes and talent management. By analyzing these shifts, researchers can better understand the evolving nature of workforce management and its implications for community service delivery. This

understanding is crucial for addressing employment inefficiencies and for developing innovative strategies to improve job matching and satisfaction among skilled labourers. Future studies can build on these insights to explore the broader impact of digital platforms or applications on employment trends and societal well-being.

Scope and Limitations of the Study

This study is a multi-platform system designed to assist skilled job workers in Los Baños City, making it easier for them to find work and for clients to hire them. It also benefits residents who need these services. This project includes the following modules:

Account Management Module

User Registration. The user can create detailed profiles, including their personal data, and attach a resume to make it easier for the employer to assess. Educational background, skills obtained, TESDA certifications, and their work experience can also be added. The Employers can review their profiles to determine the candidates if they fits their job openings. This feature greatly simplifies the hiring process.

Information Management Module

Main page. The user navigates to the main web page and can view the available jobs posted by clients. Both clients and users can post directly on the platform, specifying the job requirements, responsibilities, desired qualifications, and the rates or payment for the completed work. The platform allows companies to reach more potential candidates efficiently.

Category. This feature enables clients to organise and classify services based on specific criteria. It helps streamline the selection process by grouping similar services.

Search bar function. The system includes a search bar that lets users easily find job listings by keywords, location, and job type. This helps users find positions aligned with their skills.

Calendar and Scheduling function. The platform enables users to schedule service appointments and connect with workers through the details listed in their profiles. By offering an eye-catching interface, users can easily browse worker profiles, review qualifications, and initiate communication for various services. The system ensures efficient coordination between users and professionals, streamlining the process of finding and hiring skilled labour. Additionally, it provides a reliable means of verifying worker credentials and availability, enhancing trust and convenience for both parties.

Rating Function. The platform allows clients to rate and provide feedback on services rendered by workers. Clients can rate with stars and leave detailed comments on their experience, professionalism, and the overall quality of work. These ratings and comments enable transparency, helping other users make informed decisions while also motivating workers to maintain or improve their service standards. This feature fosters a collaborative environment, enhancing trust and accountability within the platform's ecosystem.

Recommendation System Module

Auto Recommend Service. It automatically recommends a connected service to the user's search results. This helps the user (client) find other connected services based on their search.

There are several constraints for the mobile-based system. First, the project aimed only to provide recommendations and did not include decision-making or enforcement capabilities. Second, the mobile-based system has a geographical limitation to Los Baños, Laguna. The targeted scope of this mobile-based system is to be utilized by the Local Government Unit of Los Baños. Hence, the applicability of the system in other jurisdictions was beyond the scope of this research study.

CHAPTER 2

REVIEW OF RELATED LITERATURE, STUDIES AND SYSTEMS

This chapter consists of a review of related literature and studies, both local and foreign, sourced from online reading resources, including news stories, posted personal reviews, blogs, and other trustworthy online publications. It focuses on additional works of literature that have a significant impact on the study's variables. It also discusses and connects relevant previous research to the study's conclusion.

Categorization by: Manual labour and Skilled trade

A traditional, perhaps outdated term, "blue collar" refers to a type of job done outside the office, often involving manual labour or trade skills (Rajeev Dhir, March 2025).

Manual labor

Manual labour (in Commonwealth English, manual labour in American English) or manual work is physical work done by humans. It is most literally work done with the hands (the word manual deriving from the Latin word for hand) and, by figurative extension, work done with any of the muscles and bones of the human body (Wikipedia, 2024). Manual workers are individuals who perform physical labour or tasks that require manual skills, such as construction workers, factory workers, or agricultural labourers (Safety and Health at Work, 2016). There are many jobs in this field, from construction to manufacturing. Some of these manual labour jobs are better for people who do not having as much experience or education. While many entry-level

manual labour jobs are available, there are also more technical positions that require certification or training to get started (Indeed Editorial Team, January 2025).

Non-Manual vs Manual Labour

Manual labour refers to work that primarily involves physical effort and the use of the body to perform tasks, or referred to as "blue collar", workers in these fields are often required to use tools or machinery. Still, the defining characteristic of manual labor is the reliance on physical action rather than intellectual or clerical skills.

Non-manual labour, often referred to as "white-collar" work, involves tasks that rely on intellectual, administrative, or clerical skills rather than physical effort. Non-manual labour is often associated with office work, where the primary tools are computers, phones, and other technologies rather than manual tools or machinery. Non-manual labour is generally viewed as higher-status than manual labour, largely because it is associated with education, specialize knowledge, and higher wages (Easy Sociology, October 2024).

Skilled Trade

Skilled trades are roles that require technical training. These are typically hands-on careers and differ from other professional careers. They rely on apprenticeships and on-the-job training programs to teach entry-level professionals on how to succeed in their trade roles. (indeed Editorial Team, 2025). When you hear people talking about skilled trades, you may think they're only referring to plumbers, welders and other careers in construction welders and other careers in construction and manufacturing.

What is a skilled trade today? It's a job that involves mastering a specific skill set, doing hands-on work, and yet doesn't require earning a four-year degree.

Traditionally, skills-based jobs have involved working with your hands, fixing or making something. In the past, skilled trades were considered "dirty" jobs. Even today, some are more physical and demanding than others, but that's just one of many myths about skilled trades. The skilled trades offer a lucrative alternative. You can learn the skills for most of these jobs at a community, technical, trade, or vocational school. Most of these jobs require earning a certificate, which takes 3-12 months, or an associate degree, which takes 2 years to complete (Mary Jo DiLonardo, 2023).

Artificial Intelligence

AI, or Artificial Intelligence, is generally thought to refer to "machines that respond to stimulation consistent with traditional responses from humans, given the human capacity for contemplation, judgment and intention."3 According to researchers Shubhendu and Vijay, these software systems "make decisions which normally require [a] human level of expertise" and help people anticipate problems or deal with issues as they come up.4 As such, they operate in an intentional, intelligent, and adaptive manner. Artificial intelligence algorithms are designed to make decisions, often using real-time data. AI is not a futuristic vision but rather something that is here today and is being integrated into and deployed across a variety of sectors (Darren M. W. and John R. A., 2018). Artificial intelligence (AI) is a vast technology used in the education sector. Several types of AI technology are used in education

(Nemorin et al., 2022). 2024 will see a surge in the blue-collar workforce in India, with notable growth across sectors such as construction, manufacturing, e-commerce, and agriculture. Anticipated spikes in demand are also expected from sectors such as logistics, mobility, retail, information technology, and banking and financial services. A recent report by Team Lease, a recruitment and human resources services company, estimated that the country has a blue-collar workforce of 450 million people. It is likely to drive up the demand for blue-collar jobs. As the landscape for blue-collar jobs evolves, incorporating AI into digital platforms can help these firms collect, analyse, and sort data on workers swiftly and unbiased. "When computers came, there was a fear that these machines would take away jobs. However, these new-age machines supported jobs and brought in more efficiency, productivity, and ease of doing work. Likewise, AI will help jobs evolve, which will also happen in the case of blue-collar jobs," said Richa Saxena, Associate Professor at the Institute of Management and Technology, Ghaziabad (Utkarsh Tripathi, 2024).

Artificial intelligence (AI) shapes our future more powerfully than any other invention of the past century. Anyone who does not understand it will soon feel left behind, waking up in a world full of technology that feels more and more like magic (Maini and Sabri, 2017). In the future, intelligent machines will replace or enhance human capabilities in many areas (N. H. Patil et al., 2023).

Technical Education and Skills Development Authority (TESDA)

Technical and Vocational Education and Training (TVET) prepare people for the world of work. It has played a special role in providing young

people and adults with the knowledge, skills, and competencies to improve their quality of life (Maclean 2006). One of the major development challenges for the Philippines is to prepare the youth (ages 15–24) for the workforce. The country is still facing low educational attainment, high dropout rates, and unemployment, particularly among youth. The Technical Education and Skills Development Authority (TESDA) provides national leadership for the TVET system by implementing competency-based curriculum standards, training regulations, and assessment and certification processes to ensure high-quality TVET delivery throughout the country (Kiran S. B. et al., 2024). Research Fellow at the Ateneo Policy Centre of the Ateneo School of Government examines the role of diplomas in the Philippine labour market. Her research finds that employers in the Philippines use the college diploma as a signal of higher productivity. Wage workers who possess diplomas are paid disproportionately higher than those who do not, even if both workers perform the same types of tasks. In labour economics, the wage premium is called the "sheepskin" effect (since a diploma is made of parchment paper) (Ateneo De Manila, 2017).

The Technical Education and Skills Development Authority (TESDA) regional office called for a change in image among blue-collar workers, especially the construction workers. "In the future, we will no longer call them construction workers but construction builders. If you are a builder, you are proud, with high self-esteem, adept and proficient in your skill, not only in one skill but set of skills," Tesda XI regional director Lorenzo Macapili told TIMES during the World Cafe of Opportunities event over the weekend (Julianne Suarez, 2018). Understanding and recognizing the contributions of blue-collar

workers is essential for supporting their needs and addressing the challenges they face. By prioritizing safety, offering training, addressing job security, and promoting work-life balance, organization can create a supportive and inclusive environment for blue-collar workers, ensuring their continued contributions to the workforce and economy (Refokus, 2025).

Mobile Application

Mobile application development has evolved significantly since the early 2000s, with major milestones including the introduction of Apple's iOS and Google's Android operating systems (Tracy, 2012). The history of mobile app design has seen changes in user interfaces, adaptive design, and the integration of artificial intelligence (Yasnikov, 2024). Maintenance practices play a crucial role in the success of mobile applications, and different models for scheduling and requirements management have emerged based on release history analysis (Li et al., 2014). The development process has become more accessible compared to earlier platforms, though some complexities remain (Tracy, 2012). Mobile applications now serve various purposes, including healthcare management, as demonstrated by a multi-platform system for patient medical records (Cunuhay Cuchipe et al., 2020). Agile methodologies, such as Mobile-D, are often employed in mobile app development to ensure efficient project management and favourable outcomes (Cunuhay Cuchipe et al., 2020).

Mobile application development has evolved rapidly since the introduction of iOS and Android, making it easier to create apps than in earlier systems (Tracy, 2012). The evolution of mobile apps is closely tied to

advancements in mobile technology, including hardware, software, and communication (Phongtraychack & Dolgaya, 2018). This evolution has led to the emergence of new service platforms and development tools, significantly impacting app creation and distribution (Phongtraychack & Dolgaya, 2018). Recent research has focused on improving app recommendation systems, such as the Probabilistic Evolution based Version Recommendation Model (PEVRM), which considers app version progression and addresses cold start problems (Maheswari et al., 2021). With smartphones becoming the primary handheld device for over a billion people, mobile web apps have become essential in both technical and commercial fields (Serrano et al., 2013). Understanding basic technologies is crucial for developers to keep pace with the fast-evolving mobile software landscape (Serrano et al., 2013).

Recommendation System

Recommendation systems have evolved significantly since 2000, adapting to changing user preferences and technological advancements. Recent approaches integrate matrix factorization with transformer techniques to capture evolving user preferences over time (Chen & Lee, 2024). Graph-based methods like SysEvoRecomd use Graph Evolution and Change Learning (GECL) to reconstruct evolving system networks, employing deep learning techniques such as Restricted Boltzmann Machines, Deep Belief Networks, and denoising Autoencoders (Chaturvedi & Tiwari, 2018; Chaturvedi et al., 2020). These approaches have been applied to various domains, including software, natural language, and retail markets. In e-learning, recommendation systems have proven effective in improving learner performance and managing pedagogical resources. The evolution of e-

learning recommendation systems has seen advancements in techniques, algorithms, and test data since 2000, with ongoing research addressing previously unexplored areas (Souabi et al., 2021).

Recent research has focused on developing automated recommendation systems to improve efficiency and performance. Auto-CaseRec, a novel framework, utilizes advanced hyper-parameter optimization techniques to select the best algorithm and parameters for recommendation tasks, outperforming individual algorithms with unmodified hyperparameters (Gupta, 2020; Gupta & Beel, 2020). In the automotive sector, MechService offers a database recommendation system for auto care, enhancing customer satisfaction through technician recommendations and feedback mechanisms (Sindhu et al., 2023). To address challenges in collaborative filtering, such as high dimensionality and data sparsity, autoencoders have been employed in recommendation systems. An autoencoder-based collaborative filtering method for product recommendations demonstrated promising results, achieving low Root Mean Squared Error values even with sparse datasets (Ferreira et al., 2020). These advancements in automated and deep learning-based recommendation systems aim to improve accuracy and user experience across various domains.

Categorization

Categorization is a fundamental cognitive process of grouping objects or events based on shared characteristics, essential for various tasks, including information management and survival (Gregory & Wang, 2021, 2023). It has been studied extensively, from Aristotle's time to modern

neuroscientific approaches, encompassing behavioral experiments, neuroimaging, and patient studies (Gregory & Wang, 2021, 2023). Categorization techniques are applied to various data types, including text, images, and voice, with text categorization being particularly important for managing large document collections (Mahalakshmi & Duraiswamy, 2012). In infants, categorization abilities are crucial for cognitive development, supporting memory, language use, and induction (Rakison et al., 2020). Research methods for studying infant categorisation include behavioural experiments and computational modelling (Rakison et al., 2020). Recent advancements in neuroscience have led to the development of computational models that aim to explain both cognitive and neural aspects of categorisation (Gregory & Wang, 2021, 2023).

Classification and categorization, while often used interchangeably, are distinct processes with fundamental differences (Jacob, 2004). Classification is an arbitrary and artificial tool used to structure knowledge domains, ensuring consistency and stability of meaning (Jacob, 2014). It involves systematically assigning entities to mutually exclusive classes based on established principles (Jacob, 2014). Categorization, on the other hand, is a fundamental cognitive tool for organizing, storing, and retrieving information (Jacob, 2014). It is more flexible and allows for the recognition of similarities across diverse entities (Jacob, 1991). While classification provides a stable framework, it can limit cross-disciplinary communication by imposing boundaries (Jacob, 2014). Categorization plays a crucial role in social identity formation, involving both similarity and difference (Jenkins, 2000). Understanding the distinction between classification and categorization is

essential for effective information organization and retrieval, as well as for facilitating communication across disciplines (Jacob, 1991; Jacob, 2014).

Technology Acceptance Model (TAM)

Davis (1989) introduced the Technology Acceptance Model (TAM) to explain how users adopt and accept technology. The model emphasizes Perceived Usefulness and Perceived Ease of Use as primary factors influencing user attitude and behavioral intention. Venkatesh and Davis (2000) further supported that when a system is perceived to be beneficial and easy to operate, users are more likely to accept and utilize it.

Venkatesh and Bala (2008) expanded TAM by incorporating system quality, interface design, and compatibility as additional determinants of user acceptance. Systems that are easy to navigate and work effectively across different platforms tend to produce higher satisfaction and encourage continuous use.

Synthesis of the Study

The literature discussed points to the multi-faceted nature of blue-collar work and the changing role of technology in improving the accessibility and efficiency of corresponding labour markets. These discussions point to the importance of manual labour and skilled trade jobs, the increasing use of artificial intelligence (AI), the contribution of mobile application development, and the role of educational institutions like TESDA in workforce preparation.

Blue-collar jobs, which are also linked to manual labour and crafts, continue to be a major part of the workforce. Physical labour entails arduous and physically demanding labour usually carried out by construction labourers,

manufacturing workers, and farmworkers (Wikipedia, 2024; Safety and Health at Work, 2016). As opposed to these, non-manual or white-collar jobs are typically office and intellectually or clerical in nature (Easy Sociology, 2024). Blue-collar jobs like plumbing, welding, and auto repair demand technical training but don't necessarily call for a four-year degree (DiLonardo, 2023; Indeed, 2025). Contrary to popular myths that create a negative view of these professions, they provide secure and financially rewarding career options.

Advances in technology, especially in AI and mobile app building, are now being applied in the blue-collar sectors. Artificial Intelligence refers to machine systems with the ability to react to stimuli in patterns that emulate human decision-making and judgment (Darren & John, 2018). AI integration in hiring and labor management systems has the potential to make hiring more efficient, eliminate discrimination, and make matching employees with jobs easier (Tripathi, 2024). Unlike the initial concerns, the technologies are not meant to displace employees but support the development of human capabilities and increase productivity.

Mobile app development has also quickly improved, simplifying the production of platforms addressing individual community and occupational requirements. Contemporary mobile apps have a varied range of purposes, from medicine to education and job services, and have increased accessibility thanks to improved software and design practices (Tracy, 2012; Phongtraychack & Dolgaya, 2018). These advances are applicable in the context of blue-collar work, as mobile platforms can be used as virtual intermediaries for career guidance, job placement, and skill development. In the Philippine context, the Technical Education and Skills Development

Authority (TESDA) has a crucial role in advancing technical-vocational education and training (TVET). TESDA adopts competency-based standards, training rules, and certification procedures to maintain the quality of skills development nationwide (Kiran et al., 2024; Maclean, 2006). In addition, attempts to rebrand construction workers as "builders" aim to improve their social standing and instil pride in their occupations (Suarez, 2018), part of a larger attempt to shift public opinion and make skilled trades more appealing.

The use of recommendation systems in job platforms has also been widely debated in recent research. Such systems employ sophisticated algorithms, such as matrix factorisation and deep learning, to make recommendations personal and overcome issues with data sparsity and dynamic user preferences (Chen & Lee, 2024; Sindhu et al., 2023). Such systems are especially helpful to match blue-collar workers with suitable job opportunities based on their skill levels and geolocations. Underlying these systems is the cognitive operation of categorisation, which allows for the structuring of information according to common attributes. This operation is a core component of information retrieval and recommendation algorithms (Gregory & Wang, 2023; Jacob, 2014).

Overall, the convergence of AI, mobile apps, and smart recommendation systems offers worthwhile opportunities to transform the blue-collar workforce sector. Backed by learning frameworks such as TESDA's, such technical tools can enhance the level of transparency in skills, ease job matching, and increase the overall effectiveness of labour systems. These observations form a solid basis for the present study, the purpose of

which is to examine mobile-based solutions to address employment issues of blue-collar workers in Los Baños.

Table 1. Matrix of Related Literature and Studies

Author(s)	Year	Title/Topic	Findings & Solutions
Rajeev Dhir	2025	Blue-Collar Job Definition	Defines blue-collar jobs as manual or trade-based labor outside offices → Provides foundational scope for identifying and targeting the study's workforce.
Wikipedia	2024	Manual Labor	Manual labor is physical work, typically done by hand → Establishes baseline understanding of the job types the system will support.
Safety and Health at Work	2016	Manual Workers	Outlines common jobs and worker characteristics → Supports accurate worker profiling for the platform.
Indeed Editorial Team	2025	Manual Labor Jobs	Differentiates entry-level and certified positions → Enables the system to address varying skill levels and qualifications of workers.
Easy Sociology	2024	Manual vs non-manual	Distinguishes blue-collar from white-collar work → Clarifies the study's focus and ensures the platform caters to the correct sector.
Mary Jo DiLonardo	2023	Skilled Trades Overview	Highlights pathways and modern perceptions of skilled trades → Guides integration of vocational training and certification in job matching.
Darren M. W. & John R. A.	2018	Artificial Intelligence	Defines AI as machines with human-like reasoning → Establishes theoretical base for AI-powered recommendations in the system.
Utkarsh Tripathi	2024	AI in Blue-Collar Sector	AI enhances efficiency and job matching → Justifies integrating AI for more effective worker–job pairing.
Maini & Sabri	2017	Impact of AI	AI improves human capabilities and shapes the future → Ensures long-term sustainability of AI adoption in

			employment platforms.
TESDA / Kiran et al.	2024	TVET in the Philippines	TESDA trains workers using competency-based standards → Aligns platform with national vocational training and skill validation.
Julianne Suarez	2018	TESDA Image Rebranding	Rebranding workers as "builders" improves social image → Encourages dignity and value for blue-collar workers on the platform.
Refokus	2025	Supportive Work Environment	Stresses importance of safety, training, and inclusion → Provides framework for embedding welfare and support features in the system.
Tracy	2012	Mobile App Development	Evolution of mobile OS increased app accessibility → Justifies mobile app deployment for broader user reach.
Phongtraychack & Dolgaya	2018	Mobile Tech Advancement	Mobile technology advances increase app adoption → Ensures scalability and accessibility of employment services via mobile design.
Chen & Lee	2024	Recommendation Systems	Uses transformers & matrix factorization for personalization → Basis for designing advanced job–applicant matching algorithms.
Sindhu et al.	2023	Auto Service Recommender	Feedback-based recommender systems improve matches → Suggests integrating feedback loops and profile refinement in the app.
Jacob	2014	Categorization vs Classification	Categorization is more flexible than rigid classification → Supports adaptive grouping of skills and job categories in the system.
Gregory & Wang	2021–2023	Categorization in Cognition	Categorization studies in neuroscience and behavior show flexible understanding → Reinforces conceptual design for user-friendly skill classification.
Lin, Y. & Yu, Z.	2023	Extending Technology Acceptance Model to higher-education students' use of digital academic reading tools on computers	Found that perceived ease of use, lecturer's positive response, self-efficacy significantly predict attitude/use; academic experience negatively

			predicted attitude. R^2 of model ranged 64.7-84.2%.
Venkatesh, V. & Bala, H.	2008	A Theoretical Extension of the Technology Acceptance Model: Four Longitudinal Field Studies	Demonstrated that the extended model (TAM 3) with additional constructs (e.g., experience, voluntariness, control) better explains technology adoption across contexts.

Conceptual Framework

Figure 1 shows the project's conceptual framework. It also serve as a guiding framework for the development of iServe+: A Service Management Process-Based Hiring Platform with AI-Driven Upskilling Pathways for Blue-Collar Workers. The system was built using Agile Method, following the stages such as product backlog creation, sprint planning, sprint execution, daily scrums, sprint reviews, and sprint retrospectives. After development, the system is organized into two main user modules: the Worker Modules and the Client Modules.

Worker and Client Modules work together to support the seamless interaction between the client and the workers. The system also included the Visual Studio Code, XAMPP, JavaScript, and PHP, which serve as the foundation for.

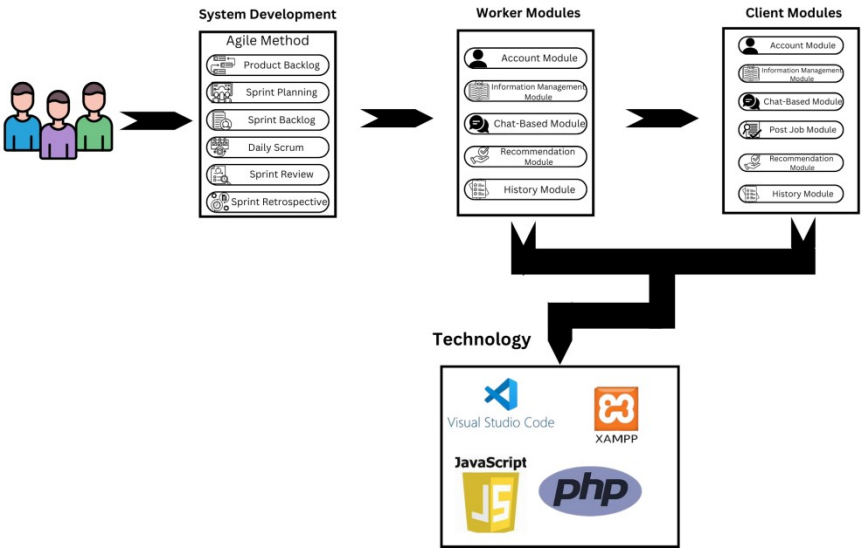


Figure 1. Conceptual Framework of the System

CHAPTER 3

METHODOLOGY

This chapter presented a systematic analysis of the methods used in the study. For the development of the application, a variety of methodologies were used as guides and a methodical approach. This chapter included the project design, project development processes, testing, and evaluation procedures.

Project Design

Figure 2 contains (2) different roles: user worker, user client. Each account or role contains modules and submodules. The client module allows searching for a service, viewing the worker's details and available schedule, rating the worker, and viewing recommended services related to the client's search. The worker module can edit their information and manage his/her schedule. The development of this system focused on easy access to skilled workers' services.

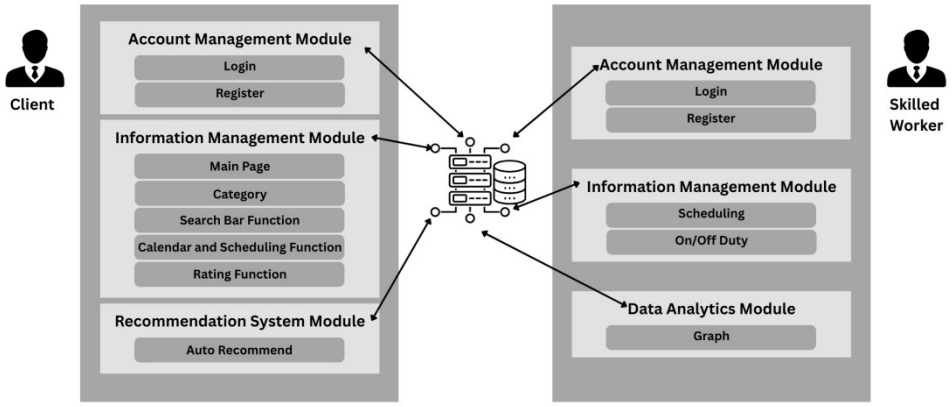


Figure 2. System Design

System Flowchart

Figures 3 and 4 illustrate the flow of the web-based system. The system started by registering an account. If the user has already created an account, they can proceed to the login page and, after logging in, access the dashboard. If the logged-in user is a worker, the system provides access only to the managed account and scheduling features. If the logged-in user is a client, they access the full dashboard. Then they can search for the services they need and also find recommended services related to the services they avail themselves of. After the user finds the services they need, they can view the details of the worker and view the schedule if they are available on the day they need the service. After the service, they can rate the service the worker provided. The client user can also manage his/her account to update his/her information.

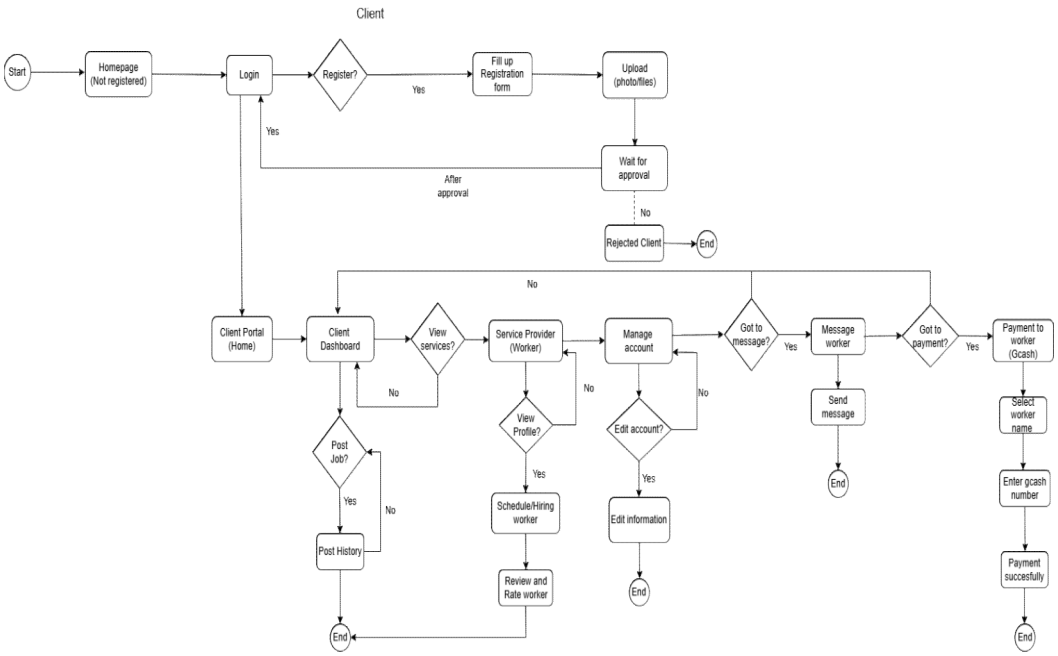


Figure 3. Shows the system flow (client)

Use case Diagram

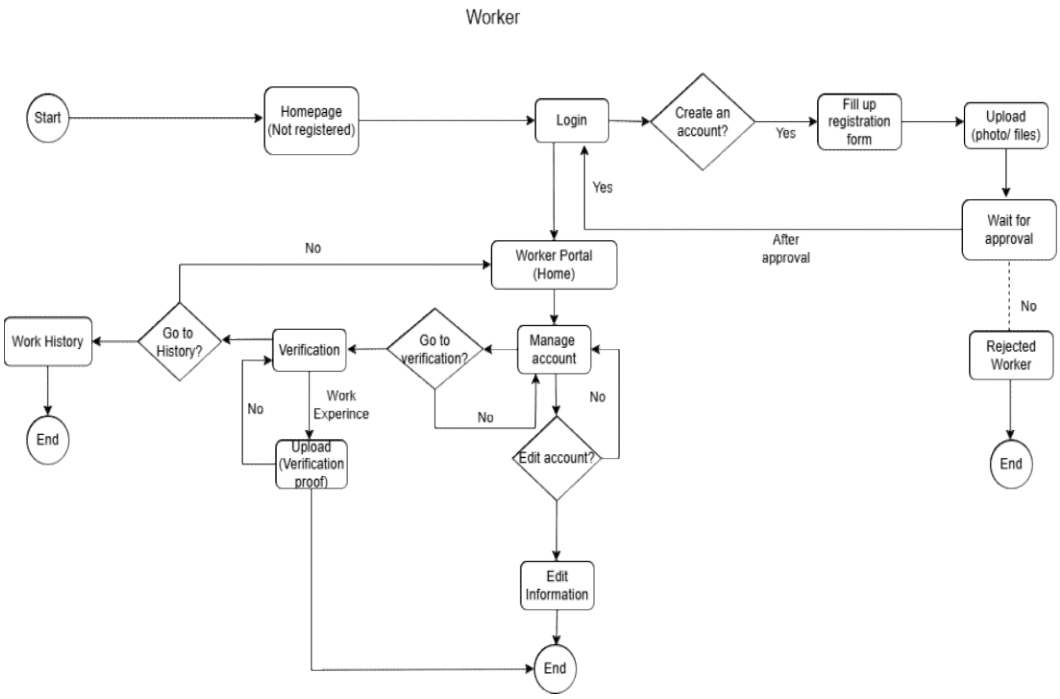


Figure 4. Shows the system flowchart (worker)

Figure 5 highlights the functional components accessible to each user type in the system, demonstrating the tailored use cases for both workers and clients. For workers, core functionalities include registering, logging in, creating and listing their services, managing their availability, and viewing their schedules to monitor upcoming bookings. This suite of tools enables workers to maintain control over their professional profiles and provide accurate, up-to-date information for prospective clients. Workers can also receive notifications for new bookings, updates, or client inquiries, ensuring seamless communication. The system allows workers to track their performance metrics, such as customer ratings and feedback, helping them identify areas for improvement.

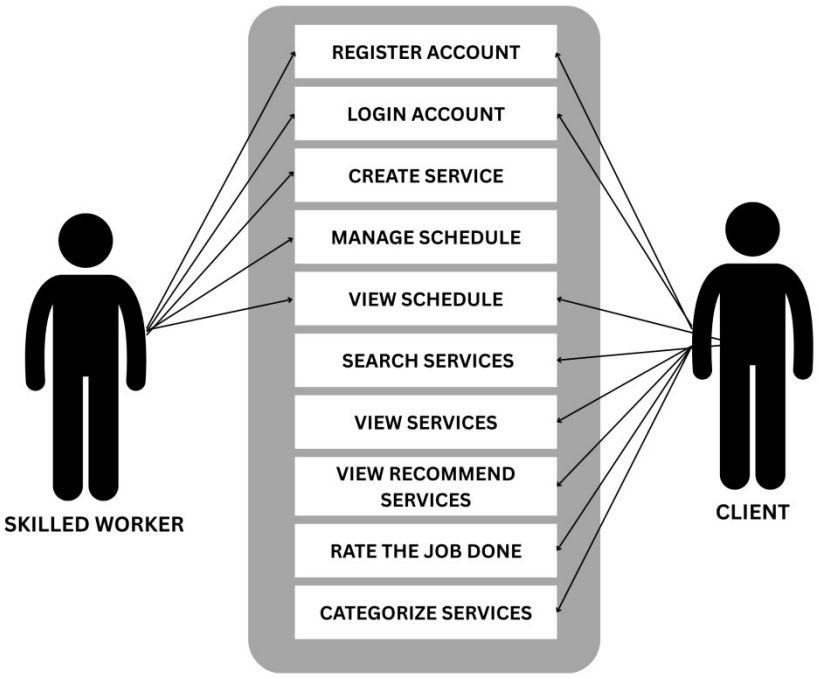


Figure 5. Use case diagram of the system

Project Locale

Figure 6 shows that the present study was conducted within the municipality of Los Baños, Laguna province, Philippines. About 63 kilometers to the southeast of Metro Manila, Los Baños is a first-class urban town renowned for educational and research institutions such as the University of the Philippines Los Baños (UPLB) and the International Rice Research Institute (IRRI). In addition to its educational and scientific nature, Los Baños also hosts a variety of labor sectors, including construction, maintenance, agriculture, transport, and services, where blue-collar jobs are essential.

It is a collaborative project with the Public Employment Service Office (PESO) of Los Baños, which is a prime local agency in the provision of employment facilitation, labor market information, and assistance to job

seekers and employers. PESO is particularly important in supporting blue-collar workers through job fairs, skills training, and employment matching.

Los Baños was chosen as the project site because of its vibrant labor market and PESO's strong presence in responding to employment demands. With this municipality as the focus, the study evaluates the current and future state of blue-collar jobs and offers recommendations to enhance PESO's initiatives to facilitate decent and sustainable blue-collar work among the local population.

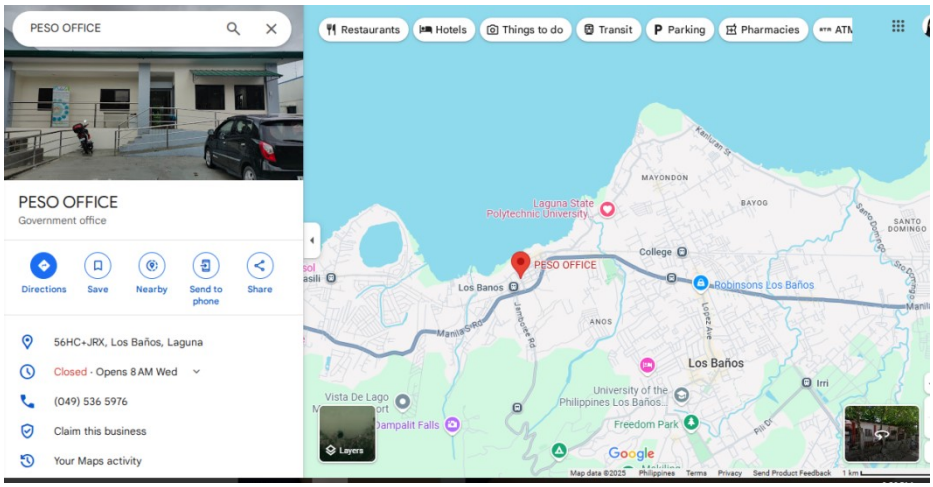


Figure 6. Project Locale

Machine Learning

The Machine Learning component of this study uses Natural Language Processing (NLP) to analyze and interpret input descriptions from both clients and workers. NLP allows the system to understand human language and extract meaningful patterns that support job matching and recommendations.

Model Description of Natural Language Processing

Natural Language Processing (NLP) was used in this research to process text-based information, including job descriptions, skill sets, work experience, and client requirements. NLP techniques enable the system to convert unstructured text data into structured numerical representations that the recommendation model can use. Specifically, tokenization, stop-word removal, and TF-IDF (Term Frequency–Inverse Document Frequency) vectorization were applied to transform text into feature vectors.

This approach is appropriate because the majority of data used in the matching process is textual, and NLP enables the system to accurately identify similarities between worker skills and job postings.

Dataset Description

The datasets used in this study were collected from clients and workers around Los Baños, Laguna. A total of 50 records were obtained, each containing textual descriptions of job roles, required skills, work experience, and applicant profiles. The dataset served as the primary input for training and testing the NLP-based matching model. Data were gathered through interviews, client submissions, and worker-provided information during registration or profiling.

Data Pre-processing

Data pre-processing was conducted to prepare the text for NLP feature extraction. The following steps were performed:

Text Cleaning. Removal of unnecessary characters, such as punctuation, numbers, and special symbols.

Lower-casing – Conversion of all text to lowercase to maintain consistency.

Tokenization – Splitting the text into individual words (tokens).

Stop-Word Removal – Filtering out common words (such as *the, is, and, etc.*) that do not contribute significant meaning.

Lemmatization / Stemming – Reducing words to their base forms (e.g., “working” → “work”).

TF-IDF Vectorization – Converting cleaned text into numerical values to serve as features for matching and similarity computation.

These pre-processing steps ensured that only meaningful information was retained for model processing.

Model Training and Similarity Matching

After pre-processing and vectorization, the system applied a text similarity algorithm to match worker profiles with job requirements. Cosine similarity was used to measure how closely the worker’s skill profile aligns with the job description. A similarity scores closer to 1.0 indicates a stronger match.

The datasets were split into 80% for training and 20% for testing to evaluate model performance and ensure that the matching process generalizes to new inputs.

Evaluation Metrics

The performance of the NLP-based matching model was evaluated using the following metrics:

Accuracy – Measures how often the system recommends the correct match.

Precision & Recall – Measure how relevant and complete the recommendations are.

F1-Score – Provides a balanced evaluation of precision and recall.

Similarity Score Analysis – Evaluates how close the worker profile is to the job description through cosine similarity values.

System Integration

The trained NLP model was integrated into the system backend. When users (workers or clients) input job or skill descriptions, the system automatically preprocesses the text, converts it to TF-IDF vectors, compares similarity scores, and generates job recommendations.

Business Model

Figure 7 shows that this project proposes a community-centered employment facilitation model titled "iServe+", designed to support PESO Los Baños in enhancing job matching and skills development for blue-collar workers. The model focuses on three key pillars: localized job profiling, employer engagement, and skills training coordination. It leverages PESO's existing infrastructure while introducing targeted innovations, including a digital job board, labor market analytics, and community-based outreach.

By aligning job seekers' competencies with the actual demands of local industries, the model aims to reduce unemployment, improve workforce quality, and foster sustainable employment pathways within the municipality.

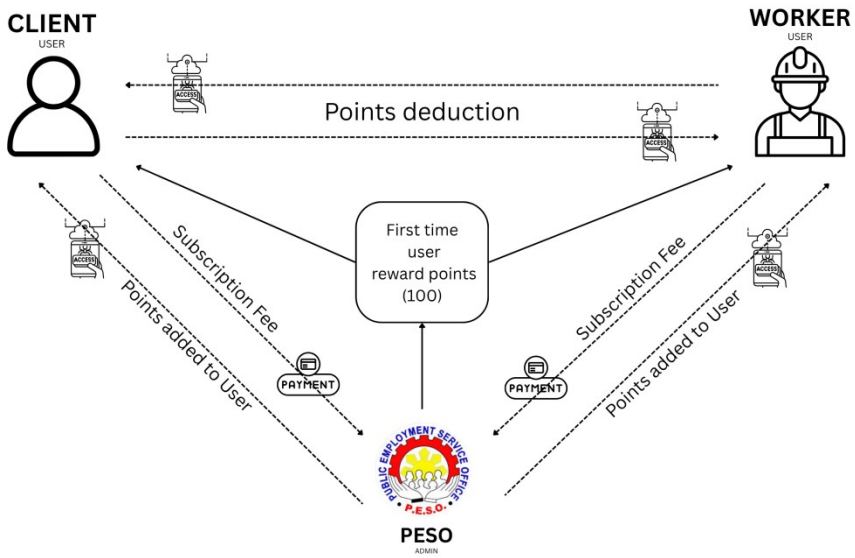


Figure 7. Business Model

Content-Based Filtering

Table 2 shows that the suggestion method for assigning employees to available jobs generally entails comparing candidate attributes, such as skills, experience, and certifications, against job specifications. Such a process often relies on algorithms that measure the similarity between candidate attributes and job descriptions, thereby increasing the accuracy of job suggestions.

Table 2. Content-Based Filtering

Worker Profile	Job Posting Requirement	Match
Skills: Plumbing	Skills Needed: Plumbing	Yes
Experience: 5 Years	Experience Needed: 2+ Years	Yes
Certification:Welding License	Certification Needed: None	Yes

After a thorough testing and evaluation process, iServe+ is officially deployed and accessible via the internet through both web and mobile applications. The system is launched with high-quality performance, ensuring reliability, user-friendliness, and efficiency for both clients and workers.

Business Start-Up for The Project

This section discusses the strategic plan for launching and growing the system, outlining primary activities and different approaches to achieve specific goals.

The iServe+ system is set to launch in 2025 and aims to provide a digital platform that connects blue-collar workers with clients who need their services. It also integrates AI-driven upskilling pathways to help workers enhance their skills and improve employability. The start-up phase focuses on building awareness, establishing partnerships, and ensuring a smooth system deployment.

The iServe+ has user-friendly interfaces that allow all users to experience convenience and efficiency, aligning with the goal of expanding the system's visibility and client base while promoting online job seeking. The

management continuously monitors the platform data and user activity to improve processes, make informed decisions, and ensure that all operations align with the platform's environment and social goals.

Marketing Plan

The project focuses on increasing awareness, building user trust, and encouraging both blue-collar workers and clients to register and use the platform. Since iServe+ integrates technology with employment support, the strategy combines digital marketing, institutional partnerships, and community outreach.

The primary market is Blue-collar workers, including electricians, plumbers, carpenters, construction workers, technicians, and household service providers. iServe+ also targets Clients or employers who require skilled workers for various services (households, small businesses, and organizations). iServe+ has a subscription fee of 30 pesos, equivalent to 100 points. It encourages users to invite others by rewarding both the referrer and the new user with loyalty points. Implement a monthly loyalty program that grants extra points to active users (up to a limit of 300 points). Collaborating with PESO and TESDA to promote iServe+ during job fairs and training programs. Work with local government units (LGUs) to include iServe+ in community employment programs. Partner with technical schools to encourage graduates to register and up-skill through

the platform.



Figure 8. SWOT Analysis

SWOT Analysis

Figure 8 outlines the strengths, weaknesses, opportunities, and threats of the iServe+ system. It provides a comprehensive framework for identifying internal and external advantages and problems, as well as external factors that can impact the system's growth. It highlights several key points to influencing the success and impact of iServe+.

iServe+ stands out for its innovative integration of artificial intelligence (AI) that personalizes skill development and training pathways for blue-collar workers. This feature enhances employability and ensures that workers remain competitive in their fields. The platform’s endorsement and collaboration with the Public Employment Service Office (PESO) also add

credibility and trustworthiness, both crucial for attracting users. Moreover, iServe+ promotes inclusivity by providing opportunities for workers who may have limited access to traditional job networks. Its reward and subscription system further encourages continuous engagement and loyalty among users, fostering a strong and active community of clients and workers.

Weaknesses

Despite its promising features, iServe+ faces several challenges in its early stages. One of the main weaknesses is the high initial development and maintenance costs required to build and sustain an AI-driven platform. Additionally, the system's effectiveness relies heavily on stable internet connectivity, which may not be readily available in rural or underdeveloped areas. Another concern is that some users, particularly blue-collar workers, may lack digital literacy, requiring orientation and training before they can fully utilize the platform.

Finally, the system's reliance on PESO's data management and coordination may pose risks of operational delays and limited scalability if institutional cooperation is inconsistent.

Opportunities

iServe+ has significant opportunities for growth and expansion. The increasing demand for digital employment platforms in the Philippines creates a favorable environment for their introduction. The system can also expand to other municipalities and regions, reaching more workers and clients while supporting local employment initiatives. Collaborations with TESDA and other government training programs present additional opportunities to integrate

formal certifications and improve worker competencies. Furthermore, iServe+’s AI-based analytics can be leveraged to forecast labor trends and identify high-demand skills, allowing the platform to stay relevant and responsive to the evolving job market.

Threats

Although iServe+'s potential is strong, it faces several external threats. The platform competes with established job-matching and freelancing applications that already have large user bases and strong brand recognition. There is also a risk of data privacy and cybersecurity issues, as the system handles sensitive user information, including personal data and financial transactions. Economic instability could affect users’ ability to consistently subscribe to or engage with the platform. Finally, iServe+ may encounter resistance from traditional employment channels, as some users and organizations may still prefer conventional hiring and job placement methods.

Materials

This section outlines the software and hardware components used in the project's development.

Software

The software used by the proponents in developing the web-based system, along with their features and the reasons for their selection.

JavaScript

JavaScript is a client-side and server-side text-based programming language that allows developers to construct interactive web pages. Whereas

HTML and CSS offer a website's structure and appearance, JavaScript adds interactive components to keep users engaged.

CSS

CSS stands for Cascading Style Sheets. It is a style sheet language used by developers to describe the look and formatting of a document written in a markup language. It provides an additional feature to HTML. It is generally used with HTML to change the style of web pages and user interfaces. It can also be used with any kind of XML documents, including plain XML, SVG, and XUL.

CSS is used alongside HTML and JavaScript on most websites to create user interfaces for web applications and many mobile applications.

Python

The system uses Python to implement the Natural Language Processing (NLP) models that power the machine learning recommendation engine. Python enables efficient processing of worker profiles, skill descriptions, and job-related text data, allowing the system to identify skill gaps, analyze similarities among worker profiles, and generate personalized upskilling recommendations.

PHP(Laravel)

The system uses PHP with the Laravel framework to handle server-side functionalities, such as connecting to and managing the database.

Laravel enhances the development process with built-in security measures, a structured codebase, and a rich library of pre-built modules. It

ensures data confidentiality and integrity for managing job profiles and matching algorithms.

XAMPP(MySQL)

XAMPP serves as the local development environment for the system, with MySQL as the chosen database. This setup enables the proponents to simulate a real-world web server environment, store job-seeker and employer data, and ensure smooth database management for features such as profile creation and job posting. CSS is used alongside HTML and JavaScript on most websites to create user interfaces for web applications and many mobile applications.

Visual Studio Code

Visual Studio Code is the primary tool for writing, updating, and debugging the system's code. Its streamlined interface supports version control, task execution, and debugging, facilitating an efficient development process for the project.

Microsoft Edge

Microsoft Edge is recommended as a testing browser because it consumes less RAM and runs faster than other browsers. This ensures smooth execution and evaluation of the system, particularly during the testing phase. Its compatibility with modern web technologies further enhances the accuracy of testing outcomes. And also, its developer tools provide robust debugging features, making it easier to identify and resolve issues efficiently.

Table 3 shows the required base details for the equipment to ensure the undertaking runs smoothly. The RAM should be 4GB to deal with the product that should have been used to promote the undertaking. A quad-core CPU is used due to the substantial software used in the undertaking. Windows 10 is strongly recommended. A computer was used during the development of the web-based system. Google Chrome is the preferred browser due to its compatibility and performance. Although the development can be conducted on various operating systems, Windows 10 is highly recommended for stability and support. Development used a computer with these specifications to ensure reliable, seamless performance throughout the creation of the web-based system.

Table 3. Hardware Requirements

MINIMUM SYSTEM REQUIREMENTS	RECOMMENDED SYSTEM REQUIREMENTS
Processor ryzen 5	Ryzen 4 above 8 gb above
Windows 10	Windows 10
Storage space 16 gb	160Gb

Project Developments

Figure 9 shows the phases of the agile scrum model. The Agile Scrum Model is used to successfully design and develop the website using an incremental process model. By employing Agile Scrum Model Development, the proponents aim to iteratively build and enhance the website until all necessary requirements are met. This approach ensures a continuous,

progressive development process, allowing researchers to closely monitor and adjust the website's evolution as needed. Agile Scrum methodology serves as a valuable framework for researchers, enabling them to efficiently manage the website development process and respond effectively to evolving project needs.

Agile Scrum Model promotes collaboration among the development team, stakeholders, and product owners through regular meetings, such as sprint planning, daily stand-ups, sprint reviews, and retrospectives. This helps ensure that tasks are clearly defined, progress is consistently tracked, and potential issues are identified early in the development cycle. Using short development sprints enables the researchers to deliver functional website components, making it easier to incorporate feedback and implement improvements. Furthermore, enhances flexibility, reduces development risks, and ensures that the final system aligns with user requirements and project objectives.

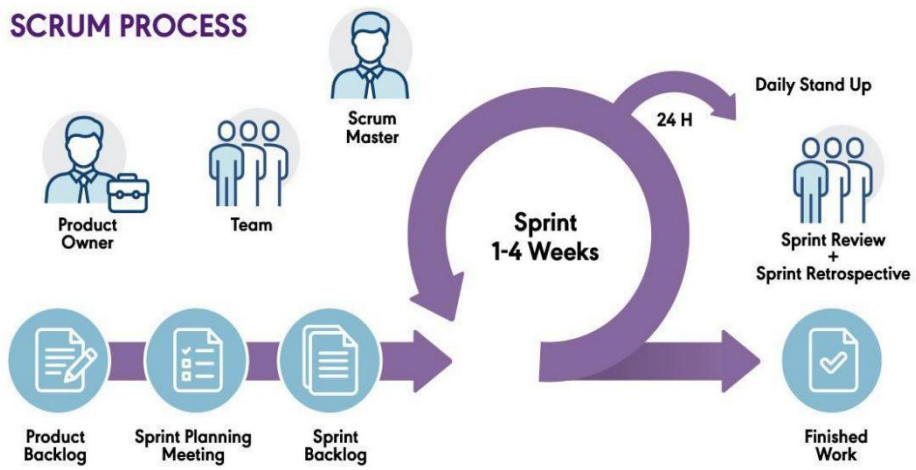


Figure 9. Agile Scrum Model

Product Backlog

Table 4 shows the product backlog that contains features and requirements for the entire system. The backlog was structured by priority, with the most critical items listed first. Additionally, the backlog included estimated hours per day, indicating the effort required for each task. This prioritized arrangement assisted in planning and ensuring that the most significant features are addressed first. By organizing the backlog in this manner, teams can effectively allocate resources and work towards the successful completion of the system. It enables developers to focus on high-value deliverables while maintaining flexibility for adjustments. The prioritization of tasks not only improves workflow efficiency but also ensures that potential risks or challenges are addressed early in the development process.

Table 4. Product Backlog for Users

Priority	Features	Estimated hours per day
	User (Worker)	
1	As a user I want to login to my account.	2hrs
2	As a user I want to edit and update my account.	4hrs
3	As a user I want to edit/update/delete a schedule from my calendar.	4hrs
	User (Client)	
4	As a user I want to login to my account.	2hrs
5	As a user I want to search for the services and workers needed.	2hrs
6	As a user I want to view the schedule of worker.	2hrs

Sprint Planning

Table 5 shows the sprint backlogs, which detail the exact items that developers planned to add during the upcoming sprint. With regard to tasks and objectives, the sprint backlogs functioned as a kind of guide for the proponents. It was also made possible for the developers to effectively and timely complete their work by including a prioritized strategy in the administrator portion of the product backlog. This approach facilitated better time management and resource allocation, enabling the team to address high-priority features first. Regular sprint reviews and updates ensured that the backlog remained aligned with project goals and client expectations. The iterative nature of sprints allowed the team to adapt to feedback and evolving requirements, enhancing the overall quality of the deliverable.

Table 5. Sprint Backlog (administrator)

Priority	Features	Task	Estimated hours per day
1	Web-based system able to login an account	Creating table in database for account. Design and develop sign in and login by programming.	2hrs
2	Add workers	Creating table in database for workers and client. Design and develop a form for adding workers and client.	5hrs
3	Create search and scheduling for workers	Programming to create search button for clients. Design and develop a calendar for scheduling workers.	4hrs
4	Rating for workers	Design and develop a rating for workers.	4hrs

Project Testing and Debugging

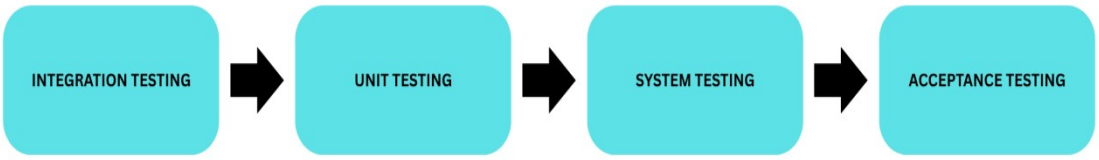


Figure 10. Project Testing and Debugging

Figure 10 shows the project testing and debugging of the system. The first phase of this system's development is unit testing, which verifies that the system is operational. This is followed by integration testing, which focuses on the system architecture's modules and includes four additional submodules: compatibility, performance, stress, and load testing. Then there is system testing, which verifies that the system is operating as intended, and finally acceptance testing, which tests the system after it has been implemented.

Unit Testing

This testing is used to determine if the website has any problems or issues. During the implementation of the capstone project, the proponents ensure that the system performs well and that the code in the software requirement analysis functions correctly to avoid errors. In this phase, the proponents test the functionality of the system, including the service search feature. They verify that job posts created by workers are accurate and that the jobs are searchable by clients.

Integration Testing

In this testing phase, the developers ensure that the modules stated in the system architecture are functioning. Integration testing is done according

to the defined system flow to verify that individual modules work together cohesively. Each integration point is meticulously evaluated to ensure data accuracy, functionality alignment, and seamless interaction among modules. Edge cases and error scenarios are tested to confirm the system's resilience and error-handling capabilities. This ensures that all modules, from user authentication to job recommendation and merit-pointing, operate as intended within the system environment. Comprehensive testing also includes validating user feedback mechanisms and ensuring a smooth user experience across various devices and platforms.

Compatibility Testing

The compatibility of each software requirement is tested, and the proponents ensure that the outcome is accurate and correct. This involves verifying that all software components, such as the database, front-end interface, and server-side functionalities, work seamlessly together. The proponents also conduct cross-platform testing to confirm that the system performs consistently across different browsers and devices. Any identified discrepancies or conflicts are addressed promptly to maintain system stability. Furthermore, stress tests are performed to evaluate the software's capability to handle high loads and concurrent user interactions, ensuring reliability under varying conditions. By conducting these tests, the proponents aim to deliver a robust and error-free system that meets user expectations.

Performance Testing

To test the performance of the system on computers and laptops, we evaluate how efficiently it operates under normal conditions. This includes

measuring the system's response time, resource usage, and overall stability during standard operations. Special attention is given to the accuracy of outputs generated by the system, such as job matches and recommendations, ensuring they meet user expectations. Performance metrics are recorded and analyzed to identify potential bottlenecks or areas for optimization.

Stress Testing

The developers push the system to its limits by simulating extreme conditions, such as handling numerous concurrent registrations or processing a large influx of job postings and applications. A critical aspect of this test is identifying the system's ability to detect and handle dummy or fake accounts effectively, ensuring robust security mechanisms are in place. Scenarios involving unusual user behaviors or malicious activities are also evaluated to strengthen the system's resilience.

Load Testing

During this phase, the proponents examine the system's ability to handle varying levels of user traffic, from light usage to peak loads. They test how quickly and efficiently data loads when users post jobs, apply for positions, or process hiring transactions. Metrics such as page load times, database query speeds, and transaction completion times are analyzed to confirm the system's scalability and consistency of user experience, even under high demand.

System Testing

In this comprehensive testing phase, the proponents assess the system's overall functionality and deployment readiness. They simulate real-

world scenarios to identify potential issues, including unexpected errors, data inconsistencies, or usability challenges. Identified issues are addressed and re-tested to ensure system robustness and reliability. Feedback from initial test users is also incorporated to further refine the system.

Acceptance Testing

This is the last phase of the project testing. The proponents break down the testing process into several phases. The first phase is unit testing, which verifies the functionality of individual code components. This is followed by integration testing, where the modules are combined and tested together. The next phase is system testing, which evaluates the complete system by combining unit and integration tests. The final phase is acceptance testing, where the system is prepared for use by users, allowing them to post and search for jobs, including opportunities for blue-collar and skilled workers.

Evaluation Process

Sampling Design

To test and evaluate the characteristics and overall performance of the system, the proponents use the purposive sampling technique, gathering and examining a total of fifty (50) target respondents. The respondents of the sampling test come from various skilled worker groups. The proponents use the Likert Scale technique to measure the system's characteristics.

Sampling Technique

The research focuses on Los Baños because the researcher also lives there, making it the most convenient location for the study. The familiarity with

the local community provides an added advantage in gathering data and tailoring the system to address the specific needs of the residents. The system would be a significant help to blue-collar workers and the broader community by providing an innovative and accessible website for matching skills with job opportunities. This project targets a diverse population, consisting of three (3) IT specialists, twenty-five (25) clients, and twenty-five (25) skilled workers. The skilled workers include individuals from various trades and professions, such as construction workers, barbers, hair stylists, manicure and pedicure technicians, and massage therapists. These groups were selected to ensure the system addresses a wide range of skill-based services and fosters meaningful connections between workers and clients.

Data Collection Instrument

In the data collection instrument for this capstone project, the ISO/IEC 25010 standard is applied, focusing on its eight quality categories: Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability. These categories provide a structured framework to assess the quality of the web-based system, ensuring it meets key standards in functionality, efficiency, compatibility, and user experience. This approach aligns well with the project's aim to create a reliable and user-centered talent-matching system.

In this project, the Technology Acceptance Model (TAM) framework is applied to assess user acceptance, incorporating a Reliability variable alongside other critical factors. Reliability is evaluated through Cronbach's alpha to determine internal consistency across seven categories of

statements. The Cronbach's alpha result of 0.778 indicates acceptable reliability. suggesting that the data collection instrument consistently measures users' perceptions, particularly in terms of system usability and user satisfaction. This level of reliability supports the credibility of the TAM analysis within the context of the project’s user-centric approach.

Statistical Treatment

In this capstone project, the proponents applied population sampling, a purposive sampling approach where the entire target respondent group was collected and analyzed to evaluate specific features and the overall performance of the system. This method was chosen to ensure that the feedback gathered was highly relevant and representative of the intended users. Additionally, the Stratified Sampling Technique was used, dividing the population into distinct groups based on respondent type, such as skilled workers, clients, and IT specialists, to provide a more comprehensive analysis across varied user perspectives. To assess system features, the Verbal Likert Scale Technique was employed, with responses scored from four (4) - Strongly Agree, three (3) - Agree, two (2)-Not Agree, to one (1) - Strongly Disagree. Responses were interpreted using a mean range: 3.25-4.00 as Strongly Agree, 2.50-3.24 as Agree, 1.75-2.49 as Not Agree, and 1.00-1.74 as Strongly Disagree. This scoring system facilitated a quantitative understanding of user satisfaction and system performance, enabling precise adjustments to be made during development.

The Functional Suitability, Performance Efficiency, Compatibility, and Usability. Reliability, Security, Maintainability, Portability. These

characteristics and their corresponding sub-characteristics serve as the fundamental criteria guiding the development and assessment of the project. They are rigorously adhered to throughout the project lifecycle to ensure that the final product meets the required standards in terms of functionality, performance, compatibility, user experience, reliability, security, ease of maintenance, and portability.

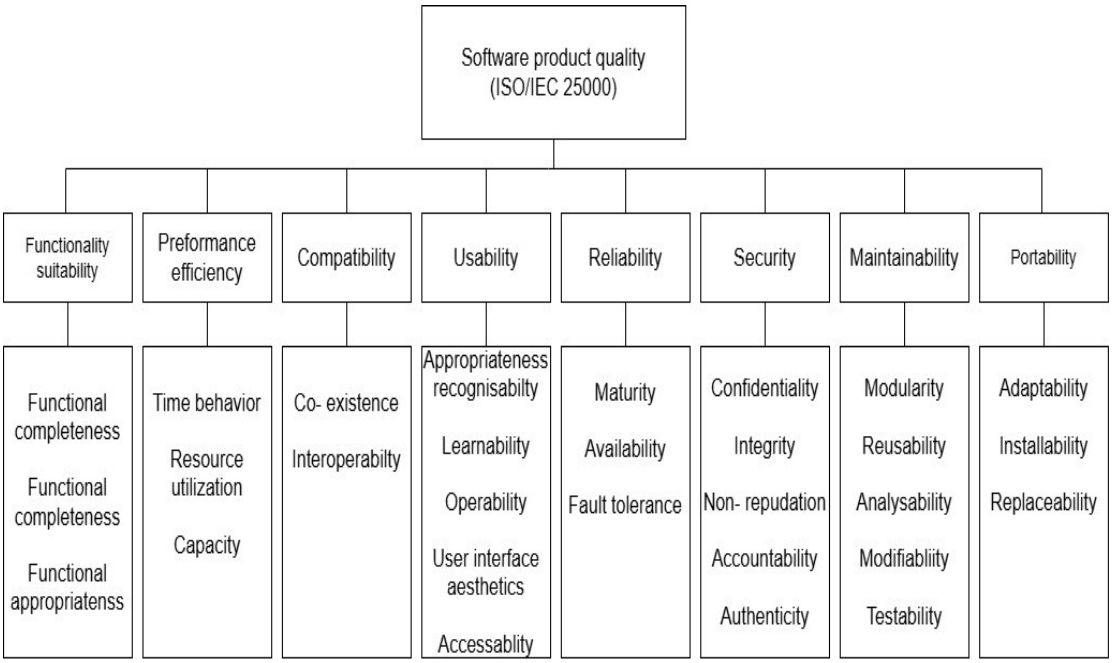


Figure 11. Quality Characteristics of ISO 25010

Technology Acceptance Model (TAM) is an information systems theory that explains how users come to accept and use a new technology. TAM highlights that actual system utilization, the point at which individuals actively use a technology, is influenced by several key factors. Traditionally, TAM identifies two primary factors that impact acceptance: (1) perceived usefulness, which reflects how beneficial the user believes the technology will be in enhancing their performance, and (2) perceived ease of use, which represents how effortless they find the technology to operate. Additional

factors also contribute to technology acceptance, including perceived cost, which considers the user's view of the technology's financial and time investment, and social influence, which reflects the impact of social networks or peers on an individual's decision to adopt.

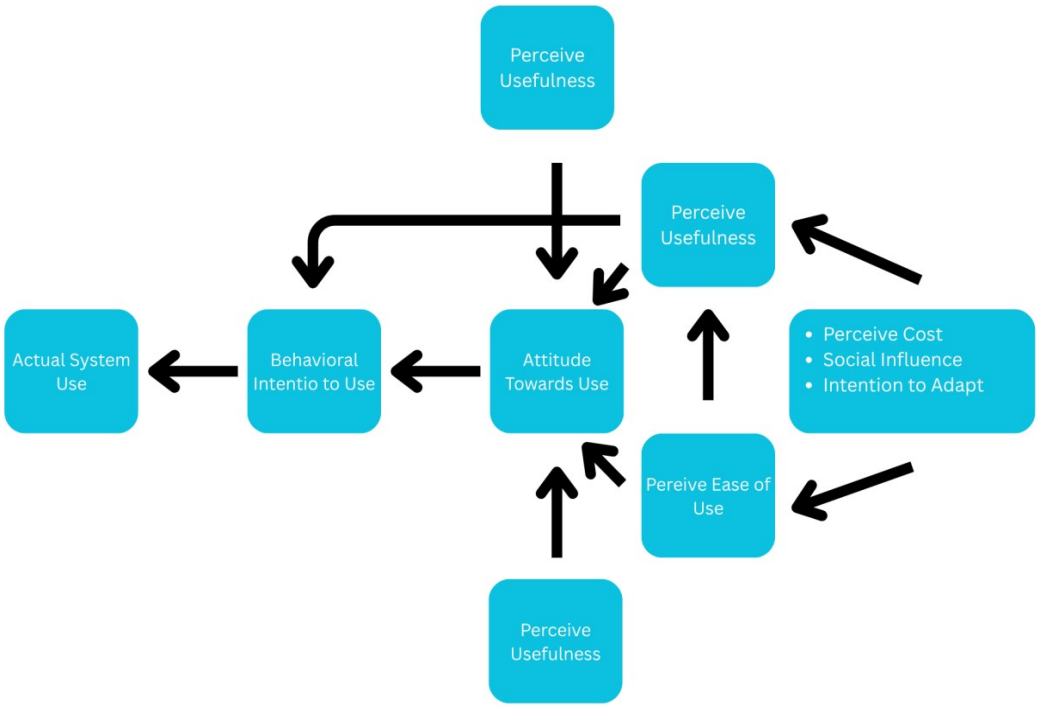


Figure 12. Quality Characteristics of Technology Acceptance Model

The model takes into account the intention to adapt and the user's readiness to integrate the technology into their daily tasks. Together, these factors shape the behavioral intention to use the technology, ultimately leading to actual system usage. This evaluation framework provides a structured way to identify and address barriers to adoption, ensuring that the technology meets user expectations and requirements. The TAM model also emphasizes iterative feedback, enabling developers to refine the system based on user perceptions and experiences. Factors such as system reliability and customization options can also enhance the perceived

usefulness and ease of use, encouraging broader acceptance. By understanding these dynamics, stakeholders can prioritize features and functions that align with user needs, increasing the likelihood of successful implementation and long-term utilization.

Table 6. The researchers used this method to determine the extent to which the system is easy to use and how each module is functional and reliable. And if the system was useful. This research was intended to collect data and responses from the conducted survey. The skilled workers and clients are evaluated by the system. As a result, it is important to evaluate the effectiveness of the system as well as how user requirements like interface design and system benefits are addressed.

Table 6. Reliability test result for Technology Acceptance Model

Categories	Cronbach's Alpha	Verbal Interpretation
Perceived Usefulness	0.74	Acceptable
Perceive Ease of Use	0.80	Good
Attitude	0.78	Acceptable
Behavior	0.61	Questionable
Perceive Cost	0.61	Questionable
Information Awareness	0.69	Questionable
Social Influence	0.73	Acceptable

Legend above .90 = Excellent, 00-085 = Good, 070-078 = Acceptable, 0.60-0.69 = Questionable, 0.50-0.59 = Poor, below 0.50 = Unacceptable

Table 7, the proponents utilized a Likert scale as the primary survey technique to measure user responses in this study. This scale provides

respondents with a range of answers, from 1 (not acceptable) to 4 (highly acceptable), allowing participants to express their level of agreement with specific statements. By using this structured approach, the proponents sought to quantify users' perceptions of the system's usefulness and ease of use, which were essential variables in understanding technology acceptance. This data was integral in assessing the system's impact and refining its functionality based on user feedback.

This assessment measures user satisfaction with the system. The proponents use a percentage-based approach, as shown in the table below, to evaluate this. This method gauged the respondents' overall appraisal of the system, helping to determine end-user satisfaction levels. The formula was as follows, Where:

The results of data gathering were encoded in MS Excel and were analyzed using the same application. The mean, median, and percentage were utilized as measures of central tendency.

Table 7. Likert Scale Technique

RATING	MEAN RANGE	INTERPRETATION
4	3.25-4.00	Strongly Agree
3	2.50-3.24	Agree
2	1.75-2.49	Disagree
1	1.00-1.74	Strongly Disagree

Ethical considerations

To guarantee that no data is shared or used for reasons unrelated to the study, the research team established strict protocols for data

confidentiality and usage. This ensures that all collected information is safeguarded and utilized solely for the purposes of the capstone project. To uphold the protection, dignity, and rights of every participant, the study strictly adheres to ethical norms and standards. The survey questions were carefully crafted, reviewed, and consulted with experts to ensure clarity, relevance, and appropriateness. All data required for the project is identified and used exclusively for research purposes. No personal or sensitive information is disclosed or employed for activities beyond the scope of the study.

To further strengthen compliance with ethical standards, the researchers aligned the study with the 2012 Data Privacy Act, ensuring participants' data privacy and security. For instance, the decision to omit barangay officials' identities was made in direct response to privacy considerations and client demands. The research team also established transparent communication with participants, clearly explaining the study's objectives, their rights, and how their information is managed. In alignment with the agreements between the researchers and the client, only information that is mutually approved is shared.

To enhance transparency, participants were informed of their right to withdraw from the study at any time without penalty or adverse consequences. Secure data storage methods, including encryption and restricted access, were implemented to prevent unauthorized usage or breaches. Researchers also conducted regular audits of data-handling procedures to ensure ongoing compliance with ethical and legal standards. In addition to consulting with legal advisors, the team sought approval from an ethics review board to validate their approach to participant confidentiality. This commitment to

ethical research practices underscores the team's dedication to fostering trust and accountability throughout the study. Additionally, informed consent forms were provided to participants, detailing the study's purpose, scope, and how their data would be used. These forms were designed in simple, accessible language to ensure participants fully understood their involvement. Regular updates were communicated to stakeholders and participants to reinforce transparency and demonstrate ethical diligence.

CHAPTER 4

RESULTS AND DISCUSSION

This chapter presents the results of the presurvey questions, provides a detailed description of the project and its significance, and includes data offering insights to help clients understand the importance and implications of this capstone project.

Results by Objectives

Web-based hiring and job-finding platform that connects employers and workers through job requirements, worker profiles, skills, availability, and rates.

Figure 13 shows the system's index page. 1) the login button for worker, 2) the login button for client, 3) the registration button for client, and 4) the registration button for worker.

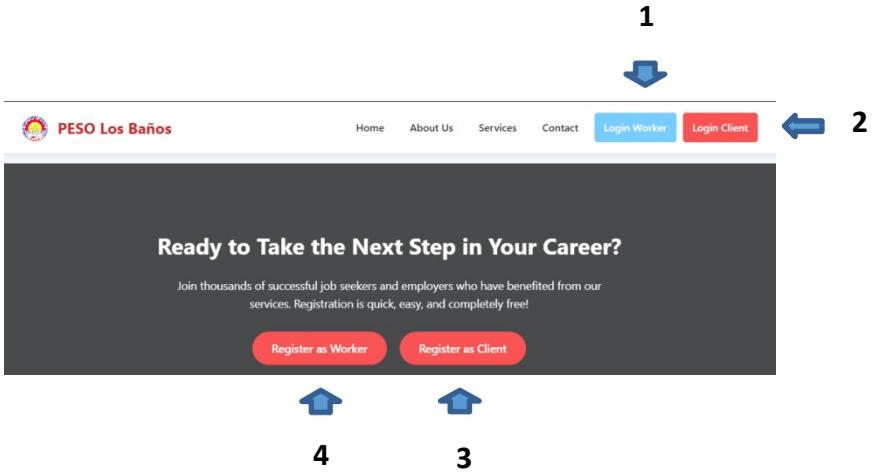


Figure number 13. Login interface of the system (for users).

Figure 14 shows the workers' registration. The system required the user (worker) to create a username using their email address and password to obtain credentials for logging in. To create the worker's profile, they need to fill out the form with their first name, middle initial, last name, date of birth, age, gender, contact number, barangay, and residential address. In skills and services, users (workers) also need to select or type their own skills, enter their other skills, and enter the service fee. Lastly, users (workers) need to provide 2x2 pictures (they can choose a file or take a photo), their resume (optional), barangay clearance, and upload any government-issued IDs.

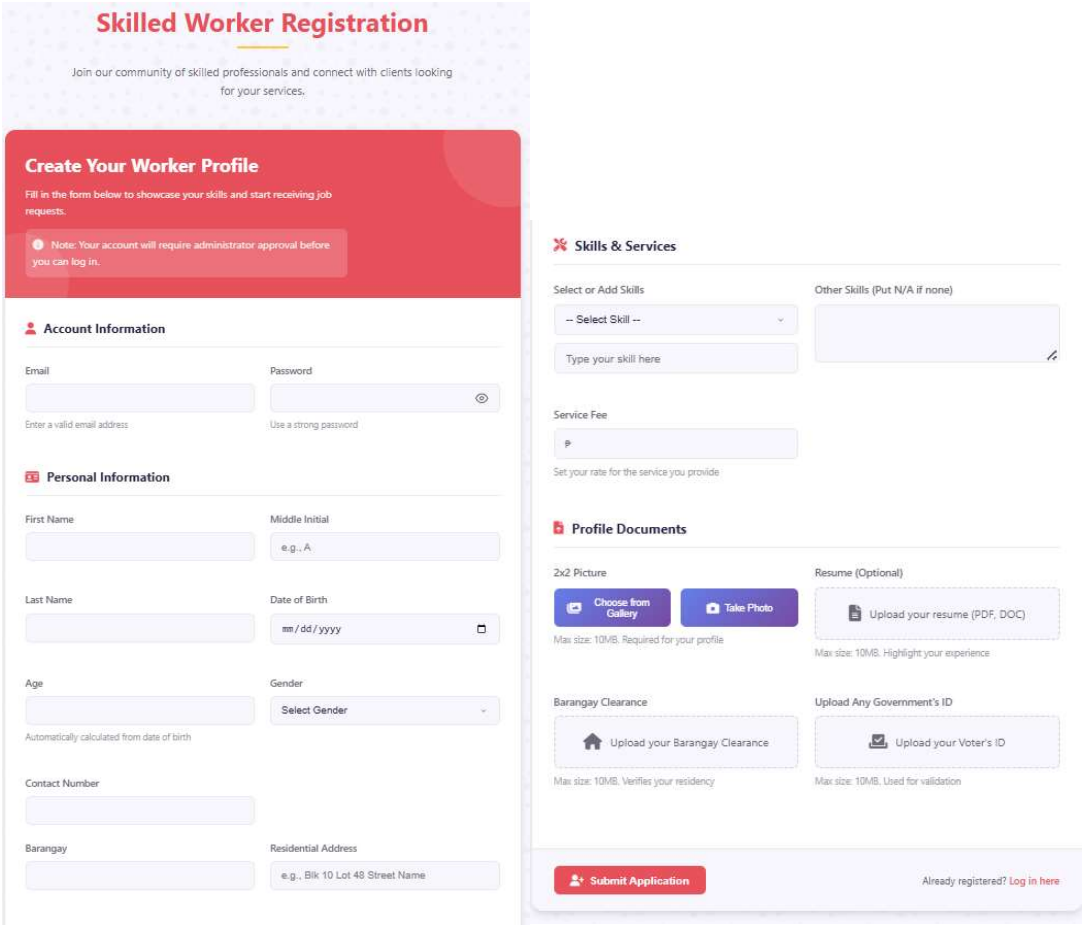


Figure 14. Registration for workers.

Figure 15 shows the client registrations. The system required the user (client) to create a username using their email account and password to obtain credentials for logging in. The user needs to fill out the form with their full name, date of birth, age, gender, barangay, contact number, residential address, and upload their profile picture. The user also needs to provide barangay clearance and any government ID.

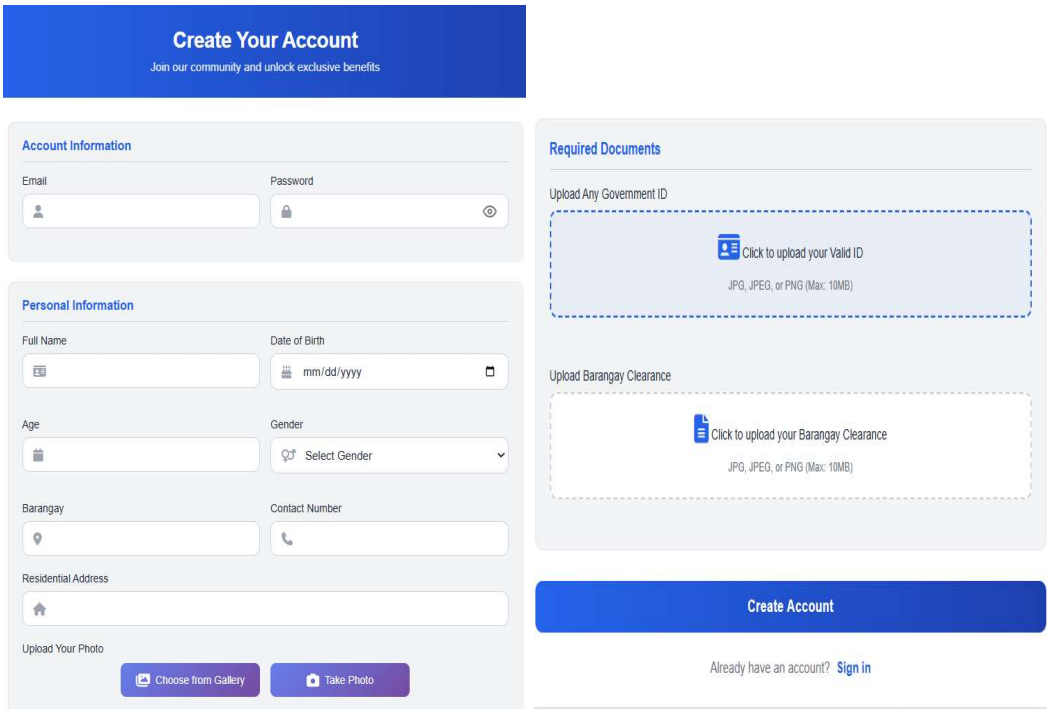


Figure 15. Registration for clients

Interface of Workers

Figure 16 shows the worker's profile page. (1) displays the user's personal details such as email address, name, age, gender, date of birth, primary skills, barangay, and contact number. (2) shows the client reviews section, where feedback from clients will appear once the worker completes jobs. On the sidebar, there are various modules including Home, Job & Training Recommendations, View Clients, Your Profile, Edit Profile, Training

Assessment Form, Upload Verification, Manage Hires, History, Messages, Buy points, and Logout.

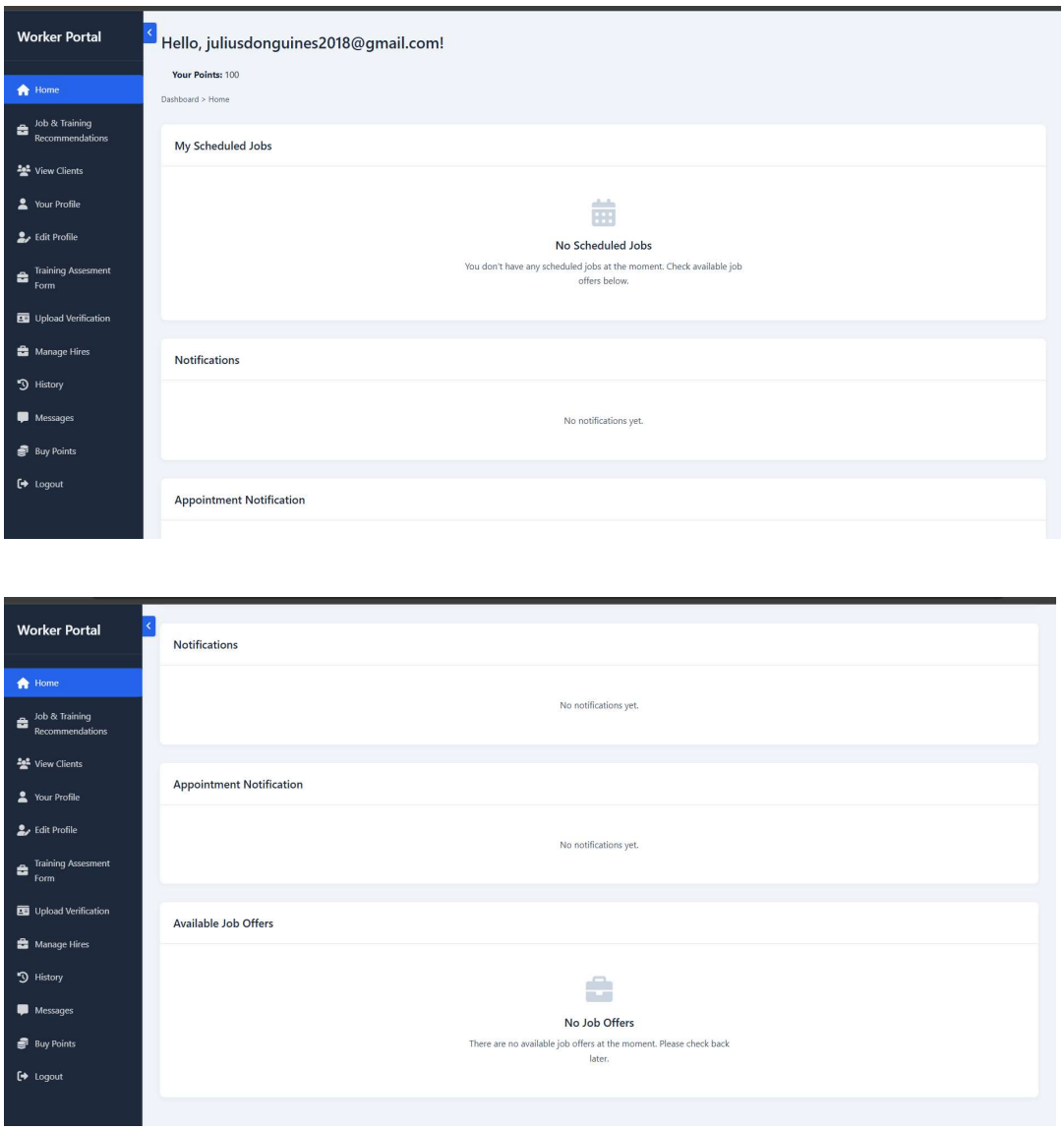


Figure16. Workers Landing Page

Figure 17 shows the profile page of the worker. (1) displays the user's personal details such as email address, name, age, gender, date of birth, primary skills, barangay, and contact number. (2) shows the client reviews section where feedback from clients will appear once the worker completes jobs. On the sidebar, there are various modules including Home, Job &

Training Recommendations, View Clients, Your Profile, Edit Profile, Training Assessment Form, Upload Verification, Manage Hires, History, Messages, Buy points and Logout.

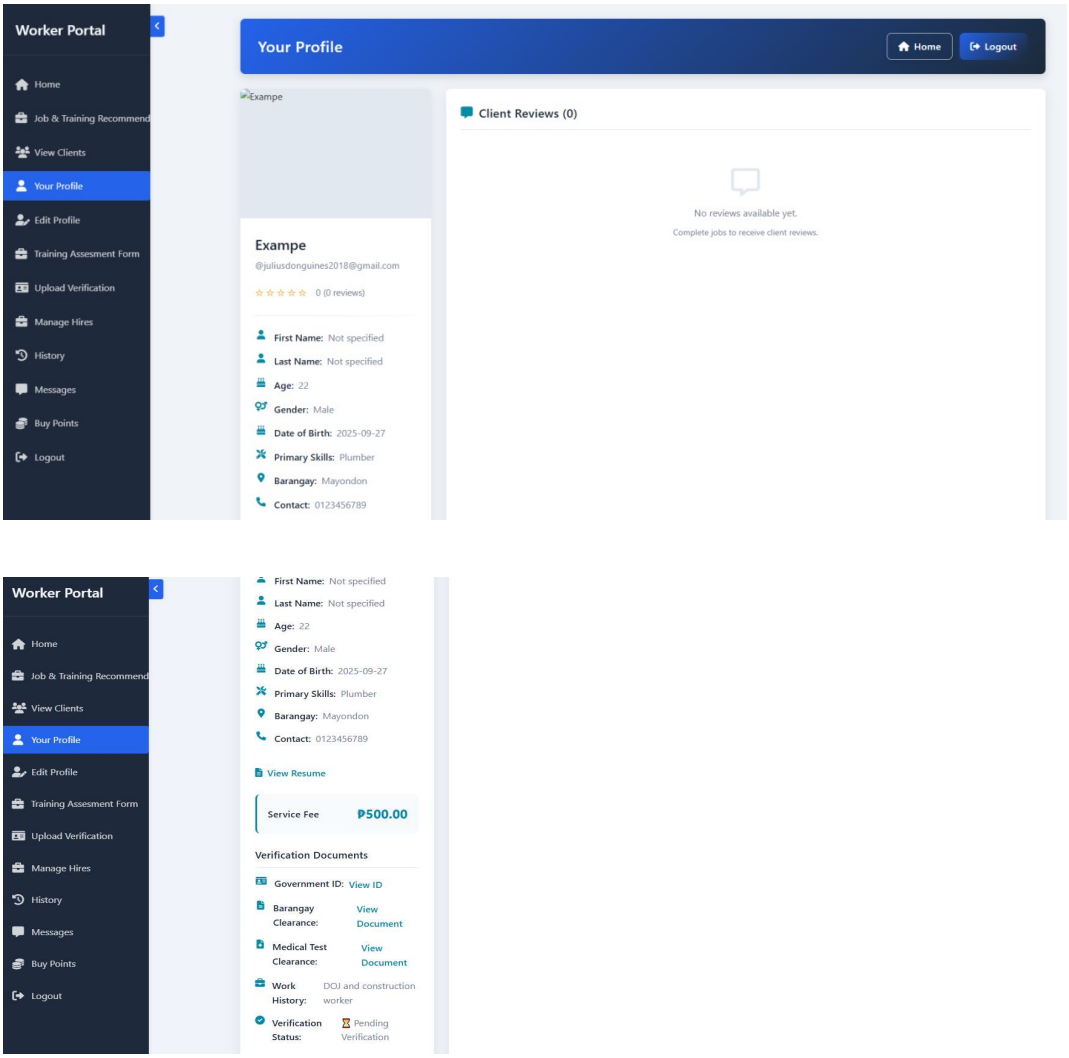


Figure17. Workers Profile

Interface of Clients

Figure 18 shows Client Dashboard Page. In the middle of the page, the client can create a job posting by entering the job title, job description, date, and time. A note informs the user that the job posting will expire if no one accepts it within 2 days. On the right-hand side, the Quick Actions panel offers

several shortcut buttons to Services, Profile, Edit Profile, and Messages. Beneath it, the Notifications section displays updates when available. At the bottom of the page, the My Scheduled Hires section lists confirmed hires, while the Post History section displays past job posts. If none exist yet, a message indicates that there are no scheduled hires or job history.

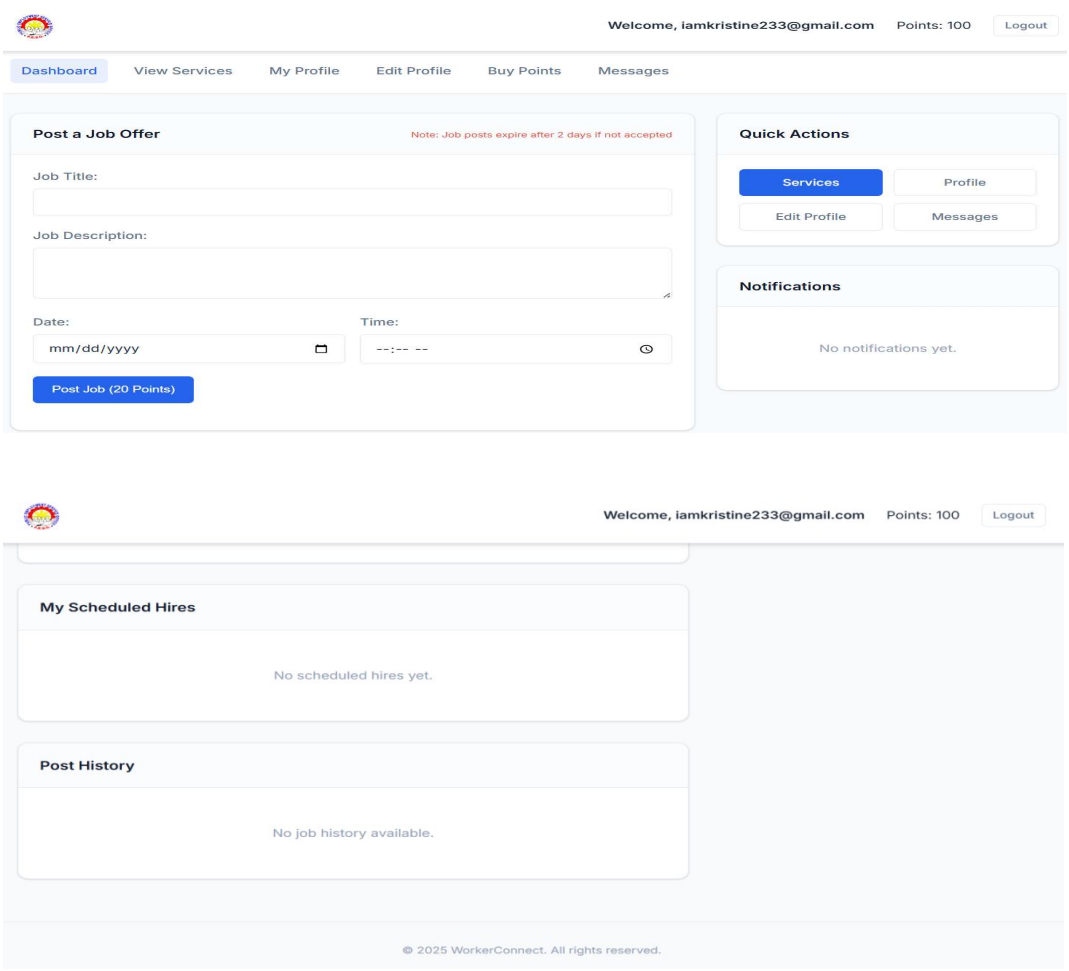


Figure 18. Dashboard of Client

Figure 19 shows View Services page, where the customer can view available skilled workers. It shows (1) each worker's profile card, showing the name, skills, other skills, location, contact information, and service fee; it also shows (2) action buttons, which include View Profile and Report. At the top, there is a Filter by Skill drop-down menu that lets the customer quickly search

for workers by skill category. This page helps clients quickly find professionals who best fit their needs.

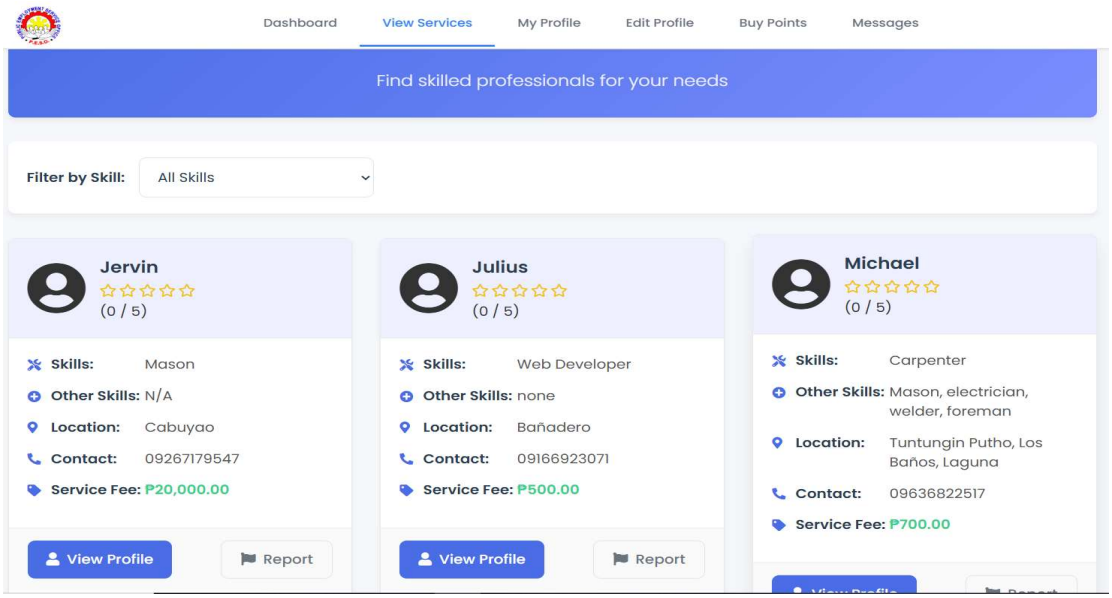


Figure 19. Available Skilled Workers

Figure 20 shows the worker's profile page. (1) displays the worker's personal information, including email address, full name, age, gender, date of birth, primary skills, barangay, and contact number. (2) shows the Client Reviews area, where feedback and ratings from clients will be shown once the worker has completed jobs. On the left sidebar, several modules are available for navigation, including: Home, Job & Training Recommendations, View Clients, Your Profile, Edit Profile, Training Assessment Form, Upload Verification, Manage Hires, History, Messages, Buy Points, and Logout.

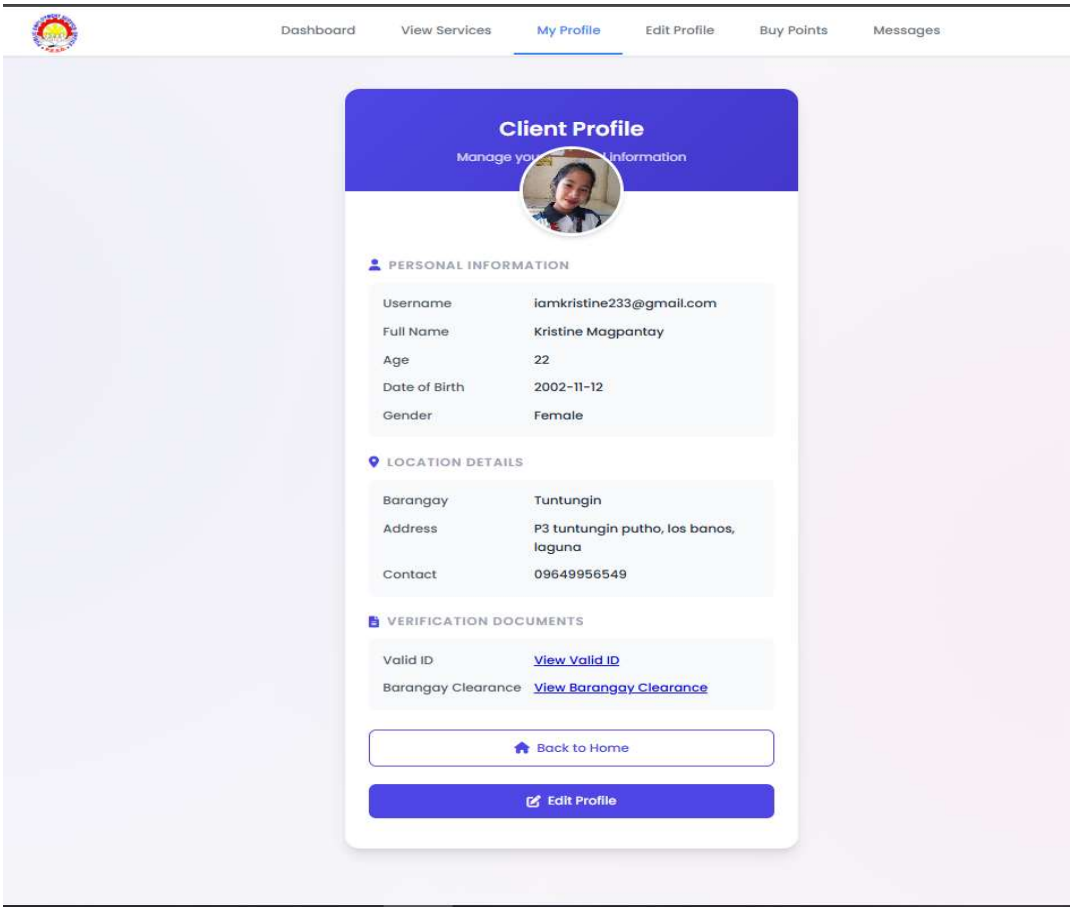
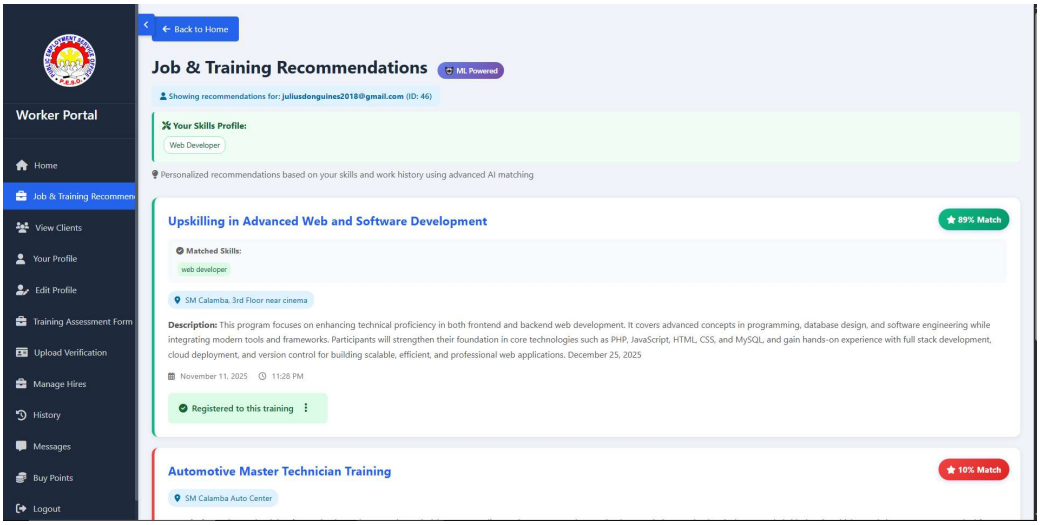


Figure 20. Clients Profile

Machine Learning-Based recommendations engine that suggests upskilling paths for workers based on market trends, feedbacks and successful profiles.

Figure 21 shows (1) the AI Career Path Recommendations section, which includes suggested career paths generated by the system, namely Senior Web Developer, Lead Web Developer, Master Web Developer, Web Developer Assistant, and Web Developer Specialist. The machine learning model generates these recommendations after analyzing workers with similar skills and backgrounds. (2) The system identifies the worker's skill gaps, such as no experience or areas for improvement, like problem-solving skills. (3) Based on these findings, the system recommends training programs with

relevance percentages corresponding to the current skill profile of the worker.was presenting the page "Job & Training Recommendations," which listed the available training programs matched to workers' skills. In this section, the match percentage produced by the ML model was included, for example, 99% Match, together with details such as a description of the training, associated skills, schedule, and registration status.



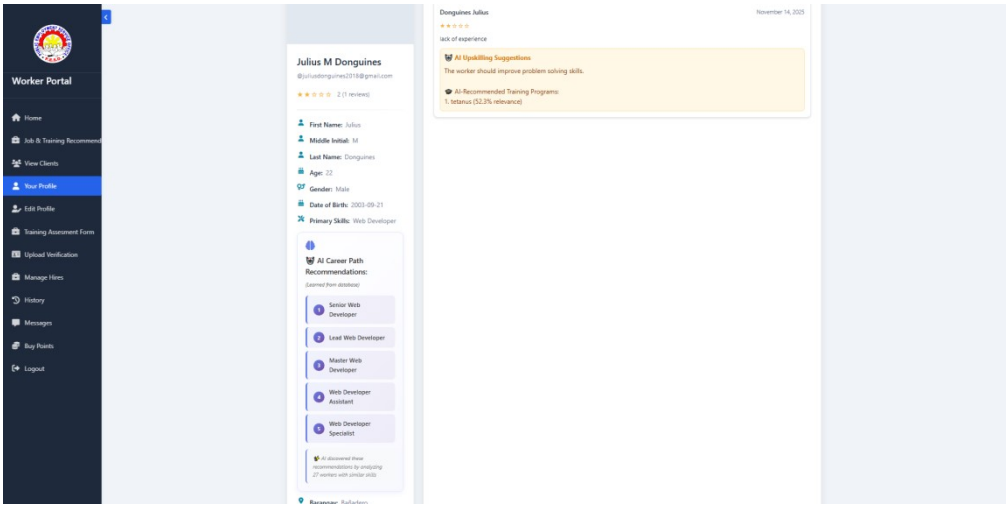


Figure 21. Machine Learning-Based recommendations engine

Result of Software Testing

The following findings were obtained from testing on the system units. The developers tested each module to ensure that the connecting modules are up to standard.

Table 8 displays the results of unit testing for the information management module, detailing the system's expected functionality, actual outputs, response times, and outcomes for each tested case. The module's unit test covers essential functions, including creating worker profiles, viewing working information, adding training schedules, and facilitating service requests. Each test confirmed the system's performance in meeting expected behaviors, with all cases achieving the anticipated results and completing within the optimal response time.

Table 8. Unit Testing for Information Management Module

Tested Case	Expected Output	Actual Output	Response Time	Remarks
Create profiles of workers	The system can create profile for the workers and put the information/detail to the system	The system successfully displays all the profile created/added to the system	6 mins	Passed
Add Training Schedule	The admin will add a training recommendation for workers.	The user(worker) successfully applied to a training recommendation made by the admin.	6 mins	Passed
View worker	The system shows the information/description of the viewed worker and all the details of the worker.	The system successfully shows the viewed details of the worker.	6mins	Passed
Request Service	The system will redirect the user(client's) to the page where they can request the service from the workers.	The user(clients) successfully redirected to the page where they can request service from workers.	6mins	Passed

Integration Testing Result

Integration testing demands the integration of all modules that contribute to the system's functionality. Connecting the market module and the data analytics module that boost the overall performance of the working system.

Table 8 shows that integration testing involves running unit tests on combined modules to ensure they work together as intended. Developers only start integration testing once each module has successfully passed functional testing. To cover all integration scenarios thoroughly, modules are tested in a

defined sequence. Additionally, selecting appropriate input data is crucial for reliable integration testing, so developers carefully choose test data for executing each test case. Integration testing helps identify issues such as incorrect data flow, interface mismatches, or unexpected interactions between modules. This stage is critical for verifying the overall system cohesion before progressing to system testing. Moreover, this testing phase ensures that data is accurately exchanged between interconnected modules without loss or corruption. Integration testing also minimizes the risk of system failure by detecting defects early, allowing developers to address issues before they escalate. Overall, this testing approach strengthens system reliability, improves performance, and ensures that all integrated components operate according to the system requirements and design specifications.

Evaluation Result of ISO 25010 and the Technology Acceptance Model.

Table 9 shows the IT specialist's overall evaluation and acceptability levels of the developed system, with a total rating of 4(GOOD), indicating that the system meets the software evaluation standard outlined in ISO 25010. This writing reflects the systems of performance across multiple criteria. Notably, the Usability and Compatibility meet the highest rating of 4.67 (EXCELLENT) of Compatibility and the total of 4.333 (EXCELLENT) of Usability, with a standard deviation of 1.15 in Usability and a standard deviation of 0.57 in Compatibility, indicating that the system's performance is perceived very positively by its users. The high mean scores demonstrate that the system is easy to understand, navigate, and operate, while also being highly compatible across the intended platforms and environments. The standard deviation values suggest that although there were slight variations in user feedback for Usability, the overall perception remains consistently favorable, reinforcing that the system successfully meets user expectations in terms of convenience, accessibility, and platform support. While functional suitability, reliability, performance efficiency, security, Compatibility, maintainability, and portability achieved the highest rating of 4 (GOOD) with no standard deviation, indicating that the system performed great across all key quality attributes. This implies that all users had a similar positive perception of the system's performance and experienced no significant issues while interacting with it. Overall, the composite mean of 4.13 (GOOD), with a standard deviation of 0.25, indicates a very good reliability among the evaluators,

underlining positively that the system performs well. A high composite mean suggests that the system effectively meets user expectations in terms of functionality, Usability, and reliability, whereas a low standard deviation shows minimal variation in responses, implying that the users had a uniform experience in their use of the system. In support of these evaluation results, the integration testing presented in Table 9 further confirms the system's readiness and functional soundness. The successful execution of the information management module and job training recommendation module demonstrates that the system's components work cohesively when integrated. The system accurately displayed worker profiles with complete service details and effectively generated job training recommendations using Natural Language Processing, with both test cases passing within an acceptable response time of six minutes.

Table 9. Integration Testing of the System

Tested case	**Expected Output**	**Actual output**	**Response time**	**Remarks**
Information management module	The system displays the workers created profiles including the service information/ detail with its fee.	The system successfully displayed all the created profiles and with its respective information.	6 mins	Passed
Job training Recommendation	The system generated and displays all the job training recommendation for upskilling in workers.	The system successfully generated and displays all the job training recommendation using Natural Language Processing.	6 mins	Passed

Table 10 presents survey results on users' perceptions of the system in terms of Perceived Usefulness, with a composite mean score of 3.55, indicating strong user agreement of its utility. The highest rated mean (mean

3.71) highlights the system's effectiveness in identifying skilled worker who meets the specific criteria, reflecting its strength in precise talent matching.

However, the lowest-rated item (mean 3.43) points to potential improvements in reducing the time and effort needed to find qualified candidates

Overall, users value the system for improving recruitment processes, aligning with research on the role of recommendation technologies. While the system demonstrates clear benefits, ongoing enhancements focused on efficiency and user feedback (client and worker) could further increase satisfaction and adoption. For instance, integrating real-time analytics or predictive features could help employers identify emerging talent trends and prepare for future needs.

Table 10. ISO25010 IT Experts Reliability Test Results

Indicates	IT1	IT2	IT3	Mean	Standard Deviation	Verbal Interpretation
Functional Suitability	4	4	4	4	0	Good
Reliability	4	4	4	4	0	Good
Performance Efficiency	4	4	4	4	0	Good
Usability	5	3	5	4.33	1.15	Excellent
Security	4	4	4	4	4	Good
Compatibility	5	4	5	4.67	0.57	Excellent
Maintainability	4	4	4	4	4	Good
Portability	4	4	4	4	4	Good

Legend: above 4.2-5.00 = Excellent, 3.40-4.19 = Good, 2.60-3.39 = Fair, 1.80-2.59 = Needs Improvement, 1.00-1.79 = Poor

Table 11 presents survey results on users' perceptions of the system in terms of Perceived Usefulness, with a composite mean score of 3.55, indicating strong user agreement of its utility. The highest rated mean (mean 3.74) highlights the system's effectiveness in identifying skilled workers who meet the specific criteria, reflecting its strength in precise talent matching. However, the lowest-rated item (mean 3.45) points to potential improvements in reducing the time and effort needed to find qualified candidates.

Overall, users value the system for improving recruitment processes. While the system demonstrates clear benefits, ongoing enhancements focused on efficiency and user feedback (client and worker) could further increase satisfaction and adoption. For instance, integrating real-time analytics or predictive features could help employers identify emerging talent trends and prepare for future needs. In addition, the highest mean scores of all indicators suggest that users perceive the system as a reliable and valuable tool for workforce management and recruitment. This overall positive response indicates that the system effectively addresses the key challenges in hiring and workforce development, particularly in supporting data-driven decision-making. Strengthening these capabilities through regular updates and incorporating user-driven improvements may help sustain high levels of perceived usefulness and ensure the system remains responsive to evolving industry demands.

Table 11. Perceived Usefulness Survey Result

Statements	Mean	Standard Deviation	Interpretation
1.The system improves the efficiency of managing the hiring process and upskilling pathways for blue-collar workers through the iServe+ platform.	3.71	±0.45	Strongly Agree
2.The system provides real-time, accurate data that enhances decision-making for both clients and blue-collar workers by improving job matching, performance tracking, and skills development.	3.57	±0.54	Strongly Agree
3.Using this system increases productivity and improves the experience of both clients and workers.	3.49	±0.61	Strongly Agree
4. The system helps prevent potential issues in workforce performance for both clients and workers.	3.43	±0.54	Strongly Agree
5. I find the system valuable in managing tasks and workers effectively.	3.53	±0.54	Strongly Agree
Composite Mean	**3.55**	**±0.46**	**Strongly Agree**

Legend 4.00 - 3.25 Strongly Agree, 3.24 - 2.50 Agree, 2.49 - 1.75 Disagree, 1.74 - 1.00 Strongly Disagree

Table 12 presents survey results on users' perceived use of the system. The composite mean of 3.46 indicates that the users find the system easy to use and integrate into their workflows. The item "I find it easy to learn how to operate the system and interpret the data it provides.

Using the system requires minimal effort or technical knowledge. It achieved the highest mean of 3.49, with a standard deviation of 0.54, suggesting strong user approval and disruption to daily operations. This shows that users feel the system is adaptable to their current practices, ensuring smooth integration. Similarly, "I am comfortable with the overall

setup and functionality of the monitoring system" scored a mean of 3.49, demonstrating ease of use without the need for significant training. This suggests that the system's design is intuitive enough to allow users to quickly become proficient without requiring extensive support. The lower mean of score 3.45, "the system helps prevent potential issues in workforce performance for both clients and workers," with a standard deviation of 0.54.

On the other hand, "I find it easy to learn how to operate the system and interpret the data it provides." received a mean of 3.42, indicating that while users generally agree with the statement, there is some room for improvement. This is further supported by the same mean of the statement "The application and dashboard make it convenient for clients and workers to monitor tasks remotely" (mean 3.42), which highlights areas where further refinement to the interface could enhance Usability. Despite these areas for improvement, the overall ratings suggest that the system is well-received and accessible, especially for users with varying levels of technological proficiency. The system's strong usability results reinforce its potential to streamline talent matching and recruitment processes, making job placement quicker and more accurate. Overall, the consistently high mean scores and low standard deviations across all indicators suggest a strong consensus among users regarding the system's ease of use. The "Strongly Agree" interpretation for all items indicates that the system effectively minimizes usability barriers and supports efficient task completion. This level of Usability is critical for sustaining user engagement and reducing resistance to technology adoption. These findings affirm that ease of use is a key strength of the system, contributing positively to its effectiveness in real-world workforce monitoring

and management environments.

Table 12. Perceive Ease of Use Survey Result

Statements	Mean	Standard Deviation	Interpretation
1. The system interface is user-friendly and easy to navigate.	3.43	±0.54	Strongly Agree
2. I find it easy to learn how to operate the system and interpret the data it provides.	3.49	±0.54	Strongly Agree
3. Using the system requires minimal effort or technical knowledge.	3.45	±0.54	Strongly Agree
4. The application and dashboard make it convenient for clients and workers to monitor tasks remotely.	3.42	±0.53	Strongly Agree
5. I am comfortable with the overall setup and functionality of the monitoring system.	3.49	±0.50	Strongly Agree
Composite Mean	**3.46**	**±0.47**	**Strongly Agree**

Legend 4.00 - 3.25 Strongly Agree, 3.24 - 2.50 Agree, 2.49 - 1.75 Disagree, 1.74 - 1.00 Strongly Disagree

Table 13 results for perceived cost have a composite mean of 3.45, which indicates that users agreed that it is reasonable and cost-effective to use the system. The highest rating was given to the statement that the system does not require excessive spending (mean = 3.51), indicating that users felt using iServe+ would not cause any financial burden. Time spent and learning effort are represented with means of 3.47 and 3.40, respectively, both considered to be reasonable in spending time using the system, yet very easy to learn. On the other hand, the lowest mean score of 3.40 is found by users. "Learning how to use iServe+ does not take too much effort or resources."

This suggests that some users may still require additional guidance or clearer on-boarding materials to fully maximize the platform's functionalities.

On the whole, users perceive that whatever cost or effort they invest in using the system is worth the benefits derived from such a system. They find iServe+ practical and, thus, worth using, especially when helping themselves in managing their tasks or finding job opportunities.

Table 13. Perceive Cost Survey Result

Statements	Mean	Standard Deviation	Interpretation
1.Using iServe+ does not require me to spend too much money (e.g., data, subscription, or service fees).	3.51	±0.64	Strongly Agree
2.The time I spend using iServe+ is reasonable compared to its benefits.	3.47	±0.54	Strongly Agree
3.Learning how to use iServe+ does not take too much effort or resources.	3.40	±0.63	Strongly Agree
4.Overall, I believe that using iServe+ is cost-effective.	3.47	±0.54	Strongly Agree
5.Any cost associated with iServe+ is justified by the value it provides.	3.40	±0.60	Strongly Agree
Composite Mean	**3.45**	**±0.41**	**Strongly Agree**

Legend 4.00 - 3.25 Strongly Agree, 3.24 - 2.50 Agree, 2.49 - 1.75 Disagree, 1.74 - 1.00 Strongly Disagree

Table 14, the composite mean of 3.43 with a standard deviation of 0.48 indicates that iServe+ is generally effective in providing information to its users. The item "The platform keeps me informed about my progress and upskilling recommendations" had the highest mean of 3.56, indicating that

users find the system helpful in guiding them on skill development and career growth.

Other aspects, such as awareness of job opportunities and new features, received ratings that were somewhat lower but still positive: 3.38 and 3.43, respectively. That is, while users do get informed, there is ample room for enhancement regarding how updates and announcements are conveyed. Generally, the findings indicate that the system supports users in staying up to date, but improving notification clarity and visibility may further enhance their experience. Moreover, the consistently positive ratings across all indicators suggest that the iServe+ platform plays a significant role in enhancing users' information awareness related to employment and skill development. The strong agreement among users reflects the platform's ability to deliver relevant and timely information that supports informed decision-making. Further improvements, such as more personalized notifications or real-time updates, could strengthen user engagement and ensure that critical information is communicated more effectively, ultimately increasing overall user satisfaction.

Table 14. Information Awareness Survey Results

Statements	Mean	Standard Deviation	Interpretation
1.iServe+ provides me with clear and updated information about available job opportunities.	3.38	±0.53	Strongly Agree
2.I am aware of the new features and services offered through iServe+.	3.43	±0.57	Strongly Agree
3.The platform keeps me informed about my progress and upskilling recommendations.	3.56	±0.62	Strongly Agree
4.iServe+ effectively communicates important announcements and updates relevant to my career.	3.43	±0.64	Strongly Agree
5.Overall, iServe+ increases my awareness of opportunities and training that match my skills.	3.42	±0.53	Strongly Agree
Composite Mean	**3.43**	**±0.48**	**Strongly Agree**

Legend 4.00 - 3.25 Strongly Agree, 3.24 - 2.50 Agree, 2.49 - 1.75 Disagree, 1.74 - 1.00 Strongly Disagree

Table 15, the composite mean is 3.49 with a standard deviation of 0.50, indicating a willingness of users to use and adopt iServe+ for their needs related to career or workforce management. The highest mean score is in relation to a regular usage intention for career development, at 3.58. This would seem to suggest a good attitude toward digital platforms taking over from traditional job-searching methods. The users were also willing to learn the upskilling features of the platform, as reflected by the means of 3.47 and

3.51, thereby indicating that they see iServe+ as useful in improving their capabilities.

Overall, users are open to adopting the system, especially when they perceive continued benefits for doing so.

Table 15. Social Influence Survey Results

Statements	Mean	Standard Deviation	Interpretation
1.I am willing to use iServe+ to find and apply for job opportunities.	3.43	±0.60	Strongly Agree
2.I am willing to learn how to use iServe+ even if it requires some time and effort.	3.47	±0.61	Strongly Agree
3.I am interested in using iServe+ to improve my skills through its upskilling features.	3.51	±0.58	Strongly Agree
4.I am ready to shift from traditional job-seeking methods to using iServe+.	3.45	±0.57	Strongly Agree
5.Overall, I am willing to adopt and regularly use iServe+ for my career development.	3.58	±0.57	Strongly Agree
Composite Mean	**3.49**	**±0.50**	**Strongly Agree**

Legend 4.00 - 3.25 Strongly Agree, 3.24 - 2.50 Agree, 2.49 - 1.75 Disagree, 1.74 - 1.00 Strongly Disagree

Table 16, a composite mean of 3.52 with a standard deviation of 0.44 indicates a favorable view of the system by users. The highest statement mean (3.49) reflects that users feel good using the system to monitor performance, indicating a good fit into their practices. The other items also

received positive ratings, indicating that users see the system as helpful and beneficial; they are also likely to recommend the system to other people. This means that, on the whole, users trust the system and appreciate its purpose and design. Also, a lowest mean of score 3.43." I believe this technology can greatly benefit the workforce and client management. "Although this statement obtained the lowest mean of 3.43, it still indicates a positive belief among users that the technology can significantly support workforce and client management, suggesting overall acceptance of the system's potential benefits.

Table 16. Attitude Survey Results

Statements	Mean	Standard Deviation	Interpretation
1.I feel positive about using the system to monitor worker performance and progress.	3.49	±0.54	Strongly Agree
2. The system fits well with my workforce management practices.	3.47	±0.61	Strongly Agree
3.I believe this technology can greatly benefit the workforce and client management.	3.43	±0.60	Strongly Agree
4. Using this system is an enjoyable and beneficial experience.	3.47	±0.58	Strongly Agree
5. I would recommend this system to others managing workers and clients.	3.47	±0.54	Strongly Agree
Composite Mean	**3.52**	**±3.52**	**Strongly Agree**

Legend 4.00 - 3.25 Strongly Agree, 3.24 - 2.50 Agree, 2.49 - 1.75 Disagree, 1.74 - 1.00 Strongly Disagree

Table 17, summarizes the user's perception of the iServe+ platform's effectiveness in talent matching and providing services. With an overall mean of 3.60 and a standard deviation of 0.38 interpreted as strongly agree, users

generally view the system favorably. The highest mean was 3.68. Based on my experience with the system, “I am willing to adopt other similar digital solutions to improve workforce management and upskilling processes." This demonstrates that the platform effectively builds user readiness for further technological adoption within employment and training systems. While the lowest mean of 3.53, "I intend to use this system regularly for monitoring workers and managing client needs." This means users see the system as helpful, but they may need more time or better features to use it regularly.

The strong behavioral intention reflected in the composite mean suggests a high likelihood of continued system usage and positive word-of-mouth among users. The willingness to recommend the iServe+ platform and to adopt similar digital solutions indicates that the platform has trust and confidence in digital workforce management tools. Enhancing system usability, streamlining workflows, and incorporating user-driven improvements may further encourage regular use and strengthen long-term engagement with the platform.

Overall, the results indicate that users have a very positive behavioral intention toward the iServe+ platform. The consistently high mean scores and low variability suggest strong acceptance, trust, and readiness to integrate the system into regular workforce management practices. This favorable perception highlights iServe+ as an effective digital solution for talent matching, service coordination, and workforce development, with strong potential for sustained use and long-term impact in employment and training systems.

Table 17. Behavior Survey Result

Statements	Mean	Standard Deviation	Interpretation
1. I intend to use this system regularly for monitoring workers and managing client needs.	3.53	±0.50	Strongly Agree
2. Clients are likely to rely on the system to manage their service workforce and connect with skilled blue-collar workers in the future.	3.58	±0.50	Strongly Agree
3.Based on my experience with the system, I am willing to adopt other similar digital solutions to improve workforce management and upskilling processes.	3.62	±0.49	Strongly Agree
4. I would suggest the iServe+ platform to others involved in blue-collar workforce management.	3.68	±0.47	Strongly Agree
5. I am committed to using this system as part of my daily monitoring routine.	3.60	±0.49	Strongly Agree
, **Composite mean**	**3.60**	**±0.38**	**Strongly Agree**

Legend 4.00 - 3.25 Strongly Agree, 3.24 - 2.50 Agree, 2.49 - 1.75 Disagree, 1.74 - 1.00 Strongly Disagree

CHAPTER 5

SUMMARY, CONCLUSIONS, AND RECOMMENDATION

This chapter summarizes the findings, draws conclusions from the observations, and offers suggestions based on them.

Summary

This study aimed to develop iServe+: A Service Management Process-Based Hiring Platform with AI-Driven Upskilling Pathways for Blue-Collar Workers, an integrated platform designed to improve the employability of blue-collar workers and streamline hiring processes for employers. The system was conceptualized to bridge the gap between skills and opportunities by providing an AI-powered environment where workers can connect with employers, showcase their skills, and receive personalized upskilling recommendations.

The researchers used a developmental research method to design and develop the system. Web technologies such as PHP, MySQL, JavaScript, and CSS were used to develop the platform. These include job and worker profile management, real-time hiring processes, and an AI-driven recommendation engine that recommends relevant upskilling pathways based on current market trends, user feedback, and successful worker profiles. The ISO 25010 Software Quality Model and the Technology Acceptance Model were used to assess the system's performance and users’ acceptance of the developed system, with a focus on criteria such as functionality, usability, reliability, and user satisfaction.

All the objectives of the study were accomplished, including developing the integrated hiring and job finding system, implementing the AI-based upskilling recommendation engine, and validating the system's acceptability through ISO 25010- and TAM-based evaluations.

Conclusion

In conclusion, the system achieves the following objectives.

1. Design and develop an integrated hiring and job-finding platform that connects employers and workers through job requirements, worker profiles, skills, availability, and rates. The project successfully developed a web-based system that provides a means for clients to post job openings and to view workers who are available and whose profiles match the required skills and schedule, while workers can view job offers that match their qualifications and availability. Similarly, the platform integrates job posting, worker profile management, and automated matching to effectively bridge the gap between employers and blue-collar workers. Testing and evaluation confirmed that all core functions—registration, profile viewing, and job recommendations—were functioning properly and meeting the intended purpose of facilitating efficient and accurate job matching.

2. Design and implement a machine learning-based recommendation engine that suggests upskilling paths for workers based on market trends, feedback, and successful profiles. The platform successfully achieved this objective by generating personalized recommendations for each worker based on their skills, background, and similarity to successful worker profiles. The platform provides AI Career Path Recommendations, suggesting roles such as Senior

Web Developer, Lead Web Developer, and Master Web Developer, based on workers with similar skills and experience. The system also offers AI Upskilling Suggestions by identifying skill gaps—such as lack of experience or the need to enhance problem-solving skills—and recommending appropriate training programs with relevance percentages. Furthermore, the Job & Training Recommendations module matches workers to available training opportunities and displays match scores (e.g., 99% Match), confirming the system's ability to align worker profiles with suitable upskilling options. These results demonstrate that the machine learning engine effectively supports workers' career development by providing accurate, data-driven training and career path recommendations.

3. Test and evaluate the acceptability of the application through a software review using ISO 25010 and the Technology Acceptance Model.

The evaluation objective was successfully achieved, as supported by the results of both the software testing and user evaluation. The assessment based on the ISO 25010 quality model conducted by IT specialists produced an overall rating of 4.13 (Good), indicating that the system met the expected standards of quality in terms of functionality, reliability, usability, efficiency, and compatibility. This means that the system performs its intended tasks accurately, maintains stability during operation, and provides an interface that is user-friendly and easy to navigate.

The Technology Acceptance Model (TAM) survey results validated the system's effectiveness from the users' perspective. Respondents strongly agreed with the system's perceived usefulness, cost-effectiveness,

information awareness, and social influence, with composite means ranging from 3.40 to 3.55. These findings imply that the system is not only functional but also widely accepted by its target users—both clients and blue-collar workers. Users expressed positive attitudes toward adopting the platform, appreciating how it simplifies job matching, improves communication, and enhances overall workforce management.

Overall, the evaluation results confirm that the developed system satisfies both technical and user-centered standards. It demonstrates high potential for practical implementation as a reliable digital tool that supports efficient workforce recruitment, monitoring, and upskilling processes within the blue-collar sector.

Recommendation

This study confirmed that the development and implementation of the *iServe+* platform can greatly assist both clients and blue-collar workers by improving the hiring process and providing AI-driven upskilling opportunities.

It also helps the community of Los Banos by providing a job that is based on experiences and can give them extra income and efficiency in managing job-related transactions. However, the researchers recognize that the system can still be improved to achieve its full potential. Thus, the researchers recommend the following:

1. Future developers may consider integrating mobile application support to improve accessibility for users with limited access to desktop devices. The machine learning model can be enhanced by incorporating real-time labor

market data and analytics to improve the accuracy of upskilling and job recommendations.

2. Conduct evaluations with a larger and more diverse group of employers and workers to further validate the system’s usability and effectiveness in various industries.

3. Continuous monitoring and improvement of data protection mechanisms should be implemented to safeguard users’ personal and employment information.

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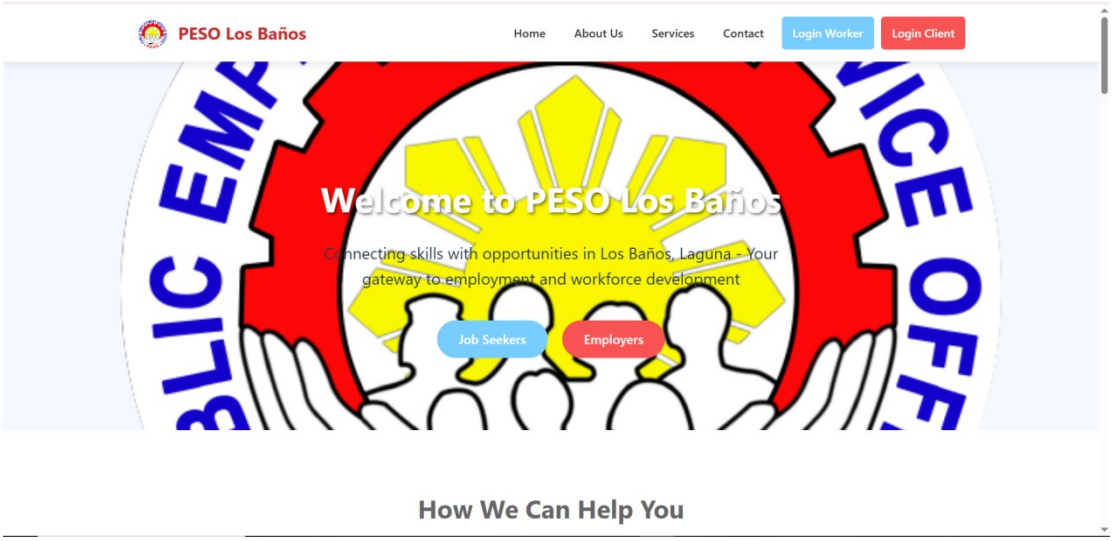
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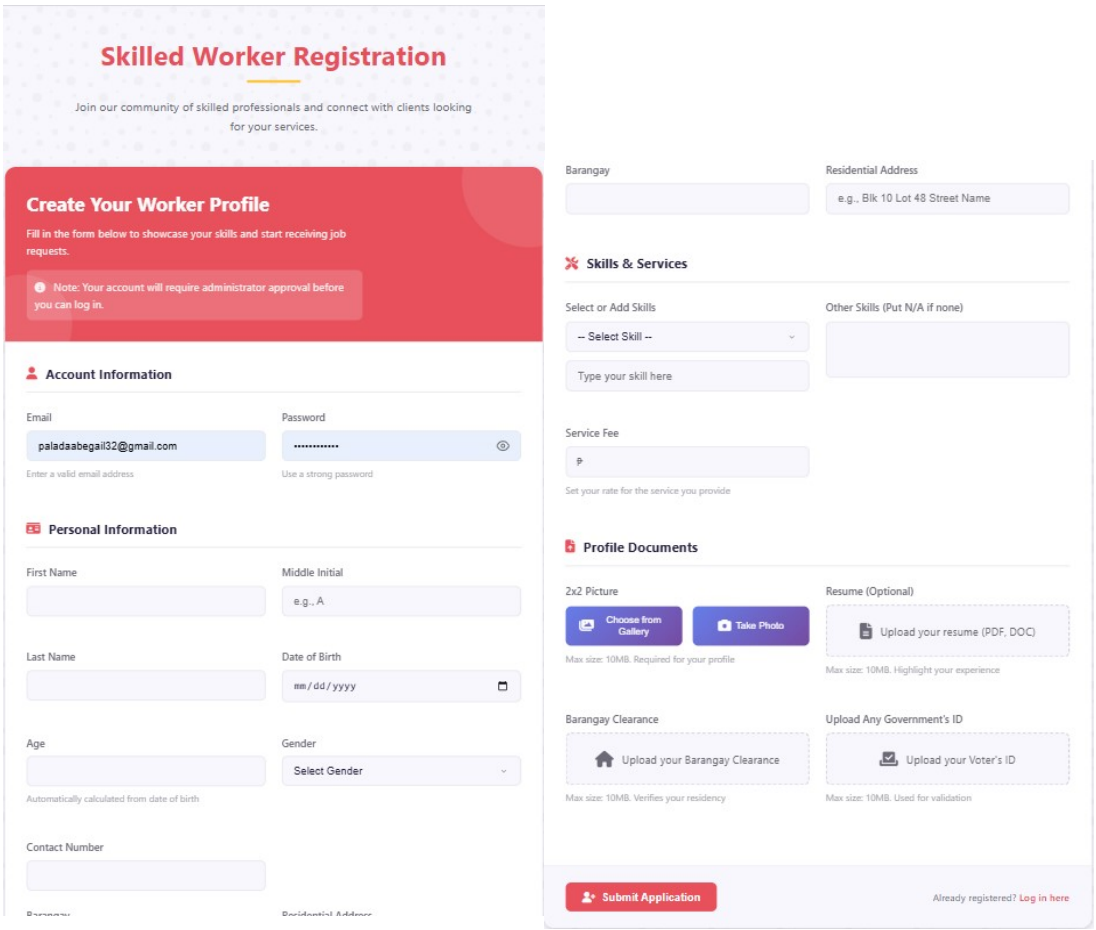
APPENDIX A

Planning & Requirement Analysis Phase

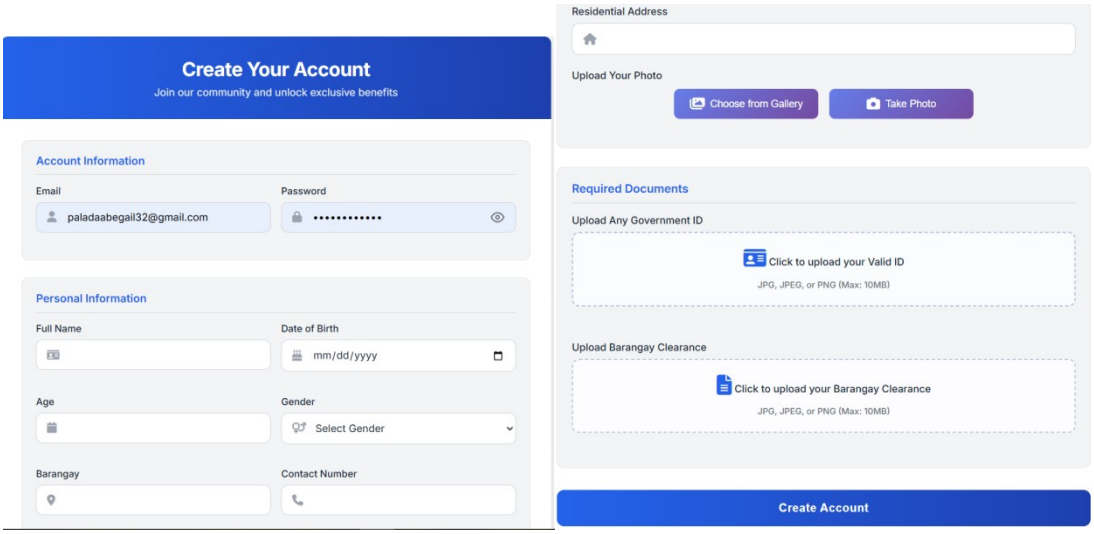
STORYBOARD



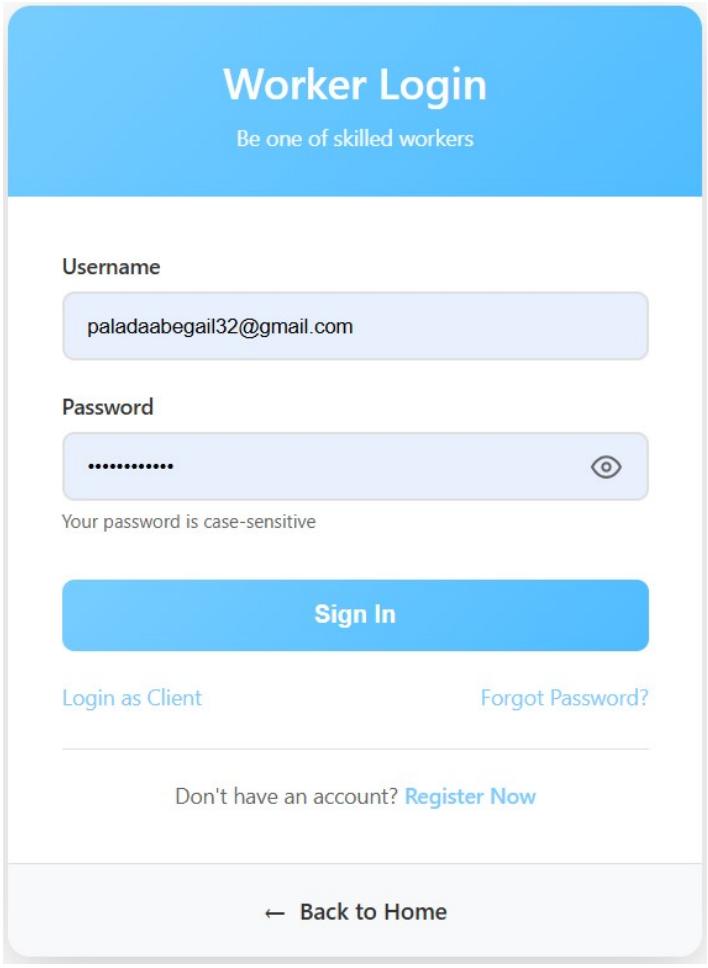
LANDING PAGE



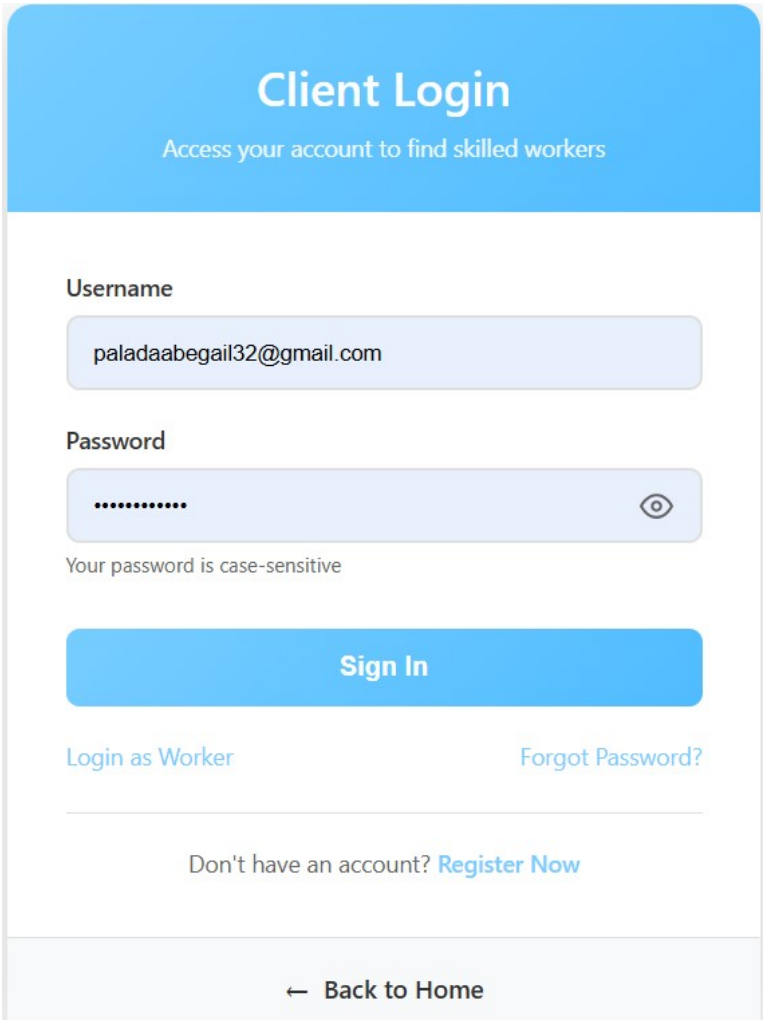
SIGN UP PAGE FOR WORKERS



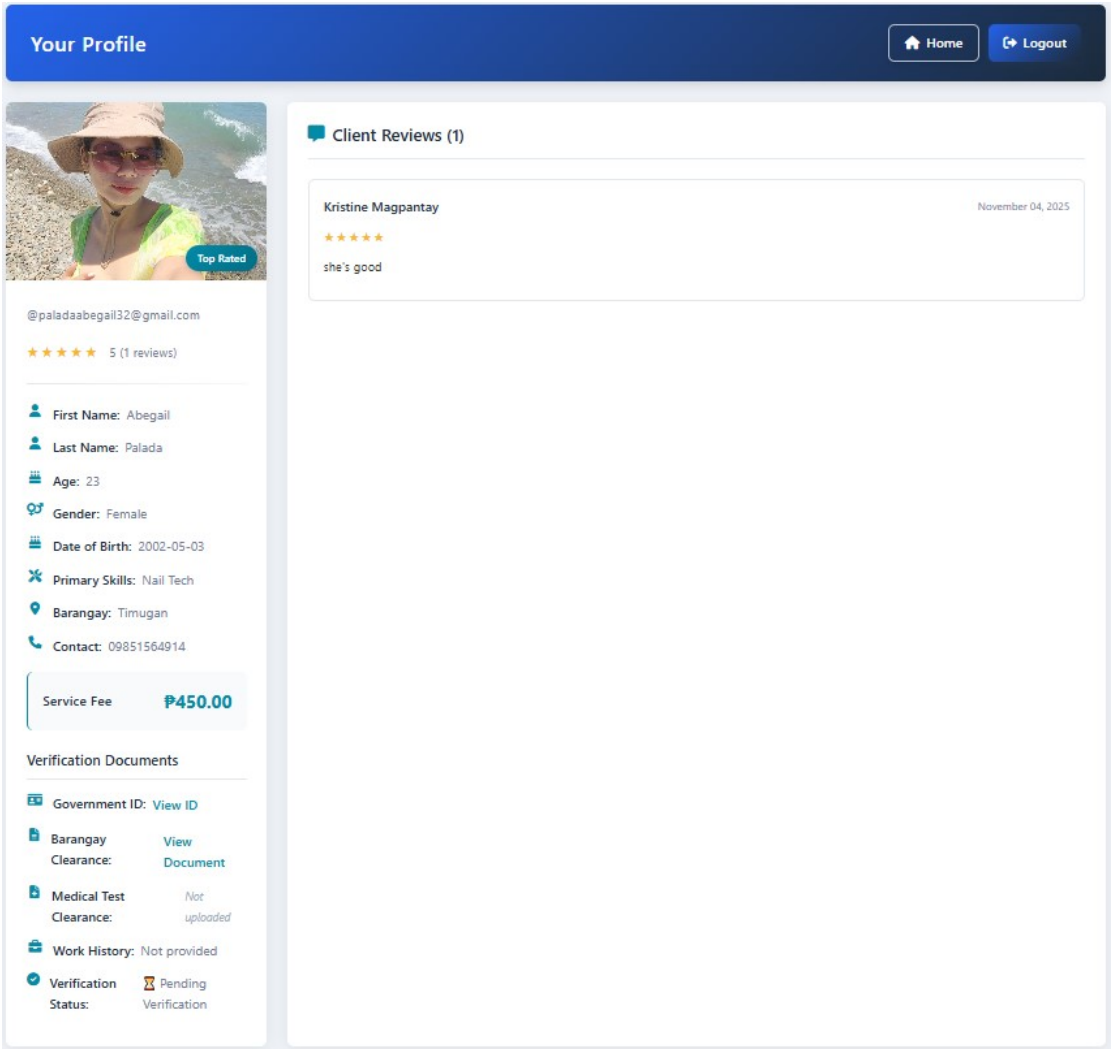
SIGN UP PAGE FOR CLIENTS



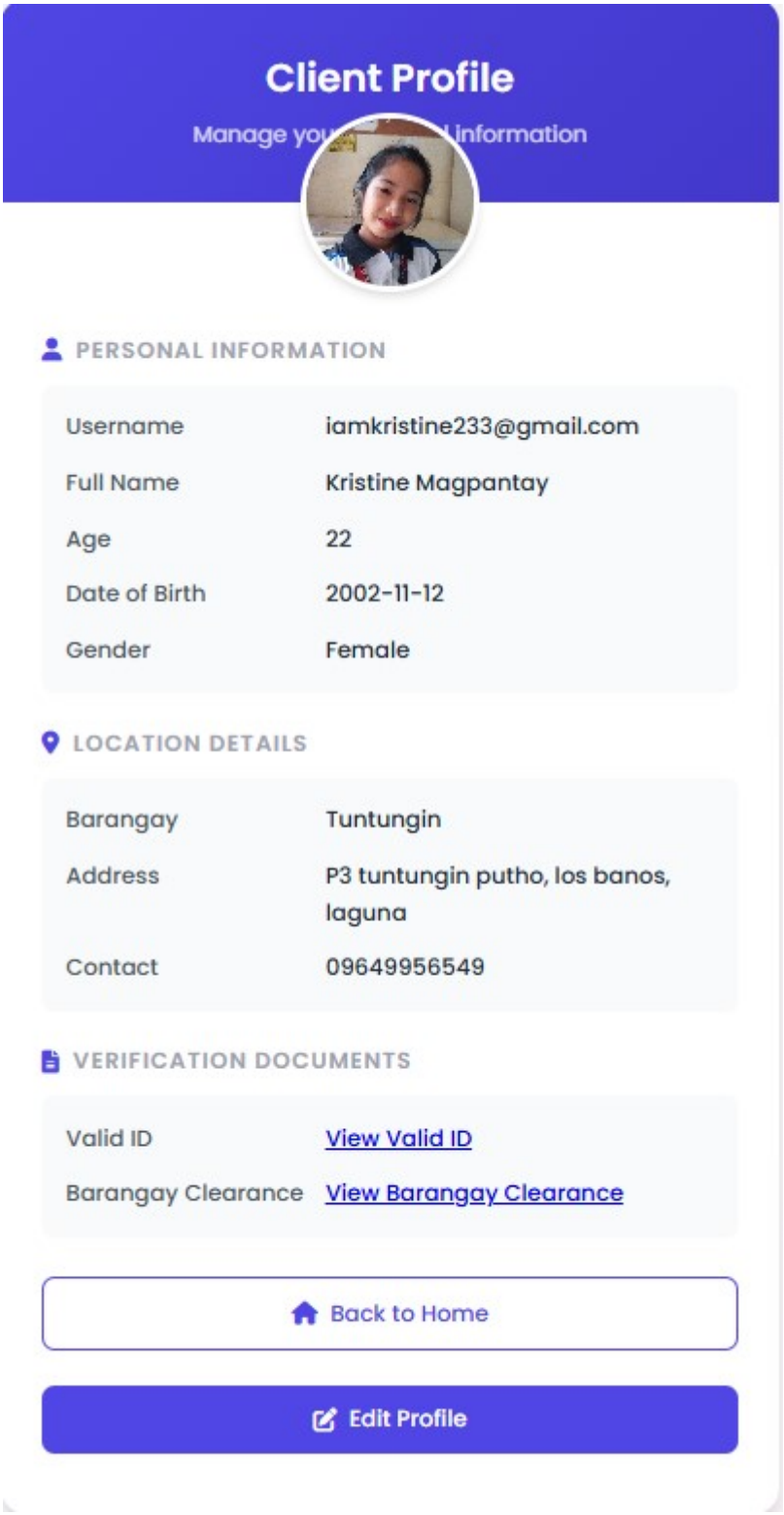
LOG IN PAGE FOR WORKERS



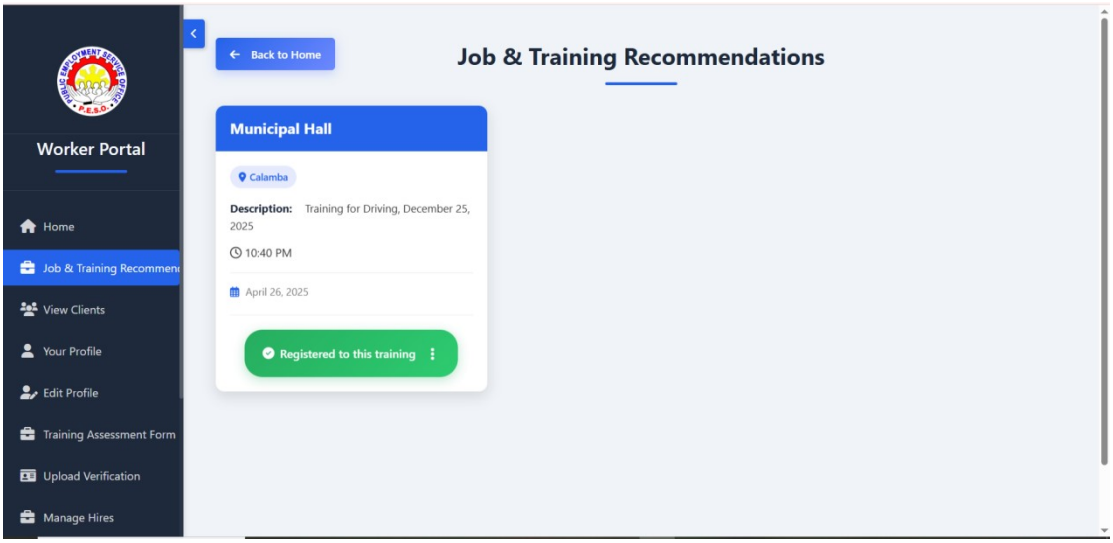
LOG IN PAGE FOR CLIENTS



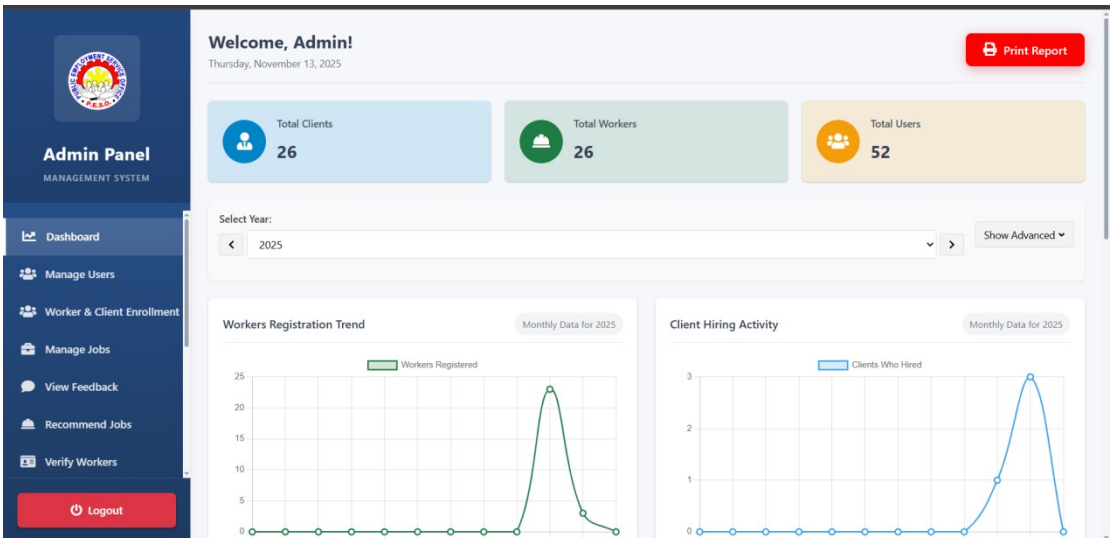
JOB SEEKER PROFILE



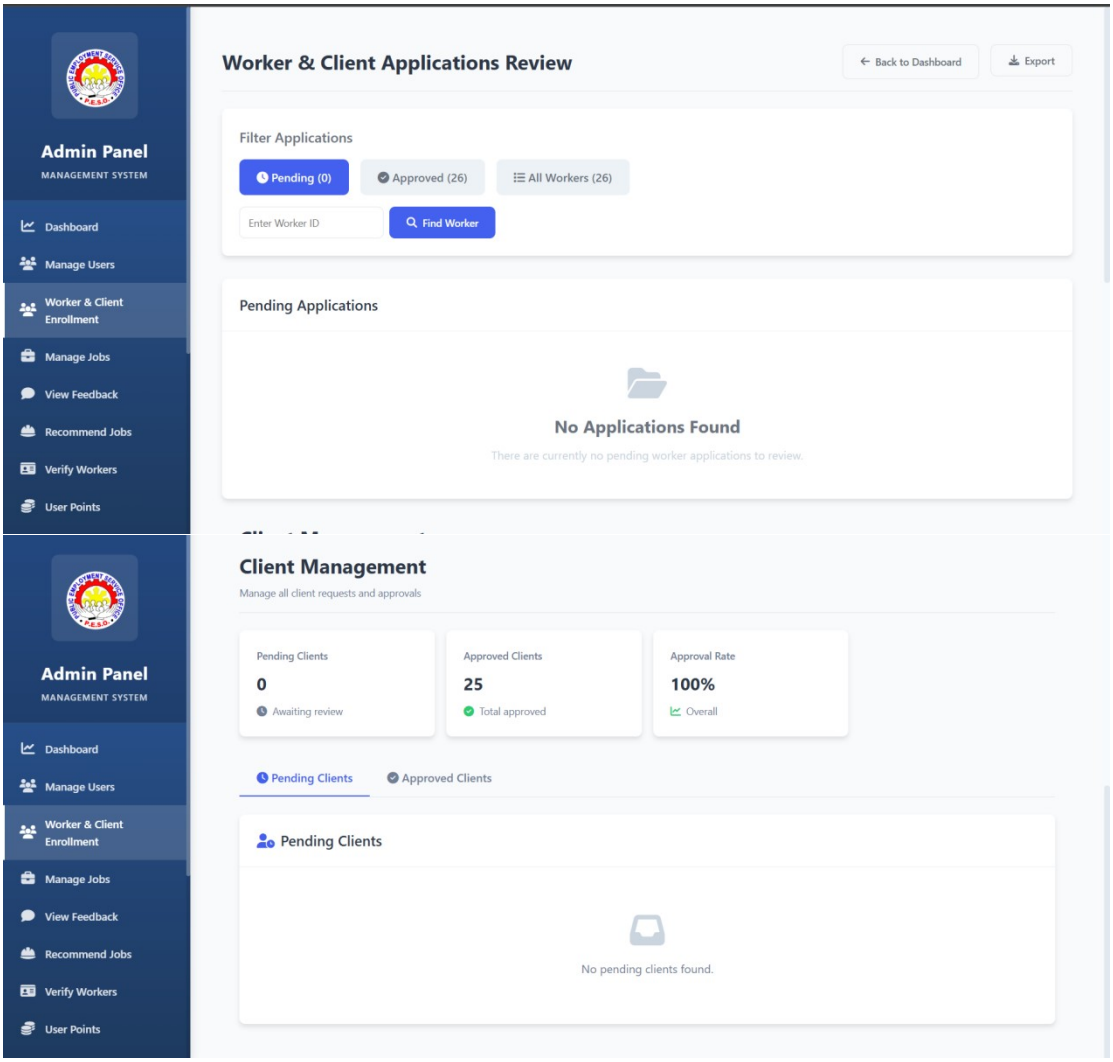
CLIENTS PROFILE



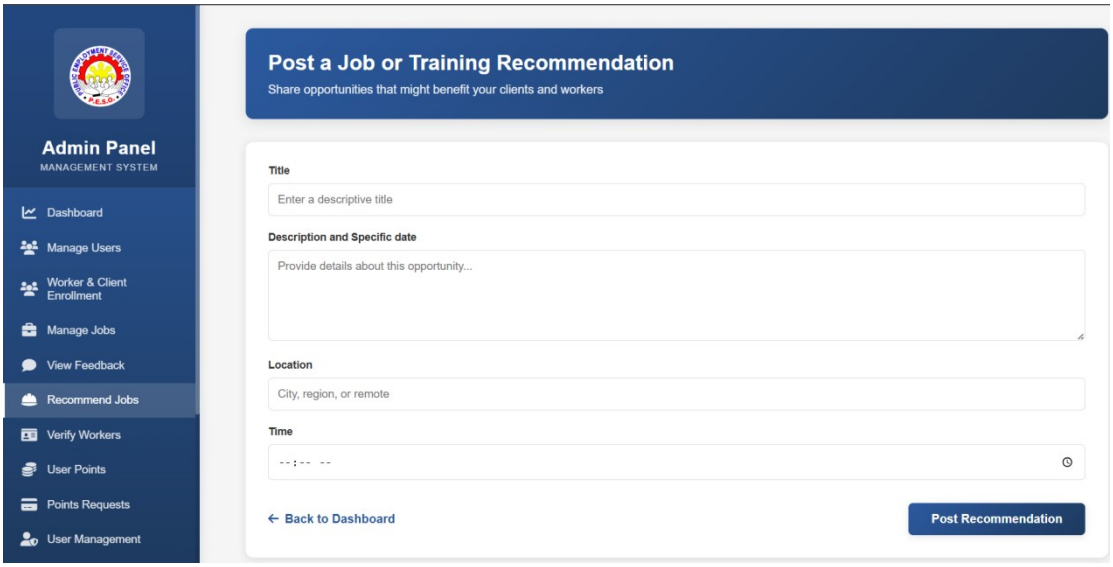
JOB TRAINING & RECOMMENDATION FOR WORKERS



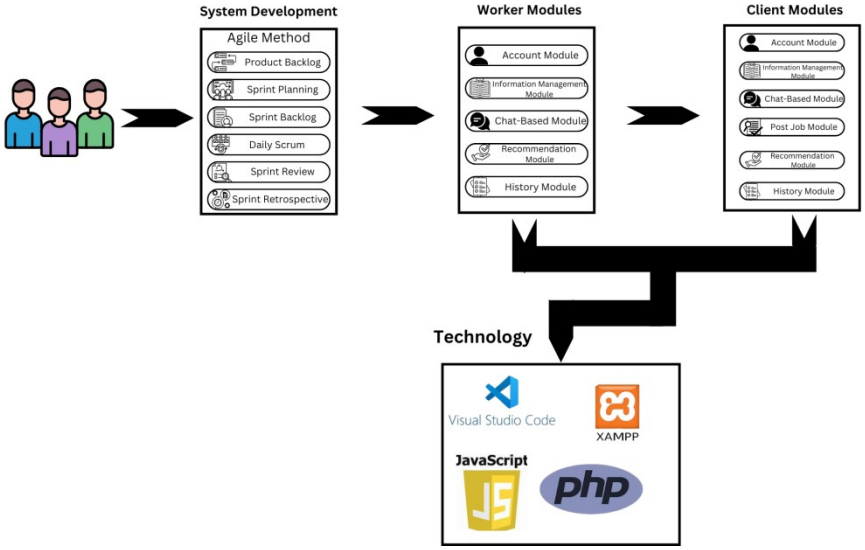
ADMIN DASHBOARD



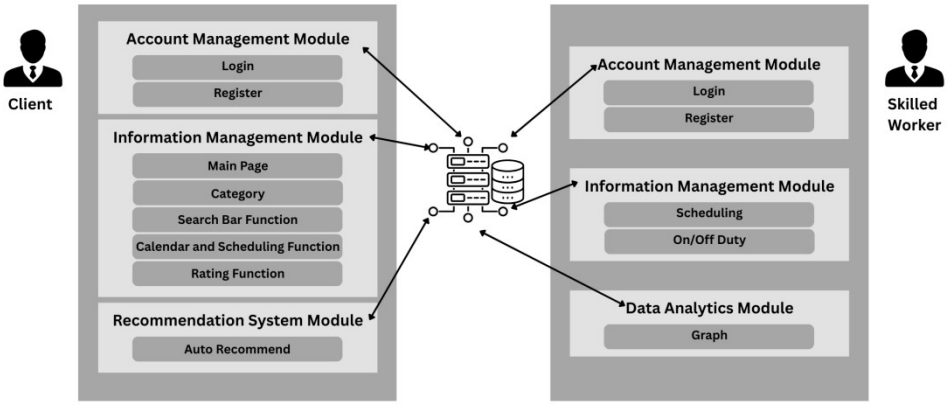
ADMIN USER APPROVAL



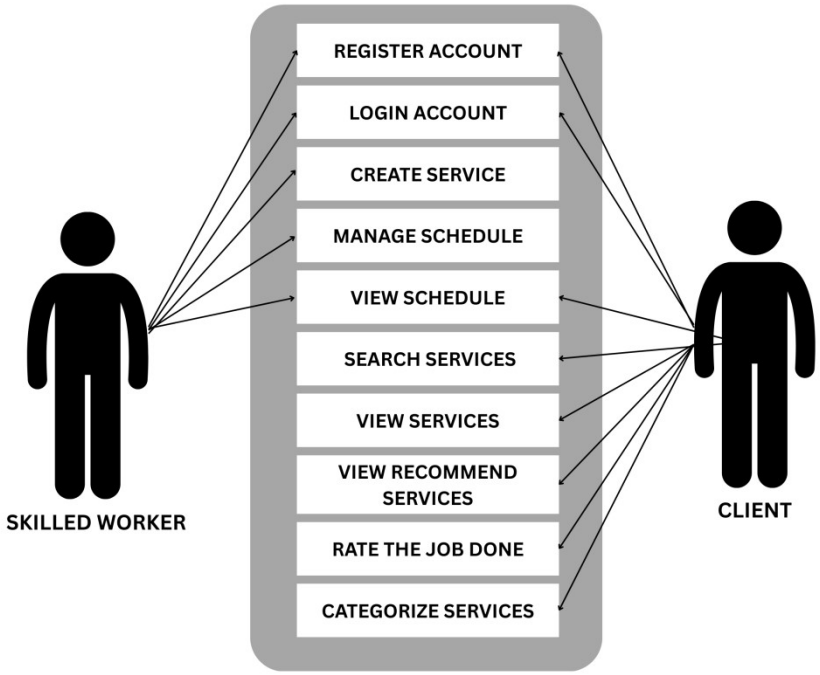
ADMIN POSTING NEW JOB & TRAINING



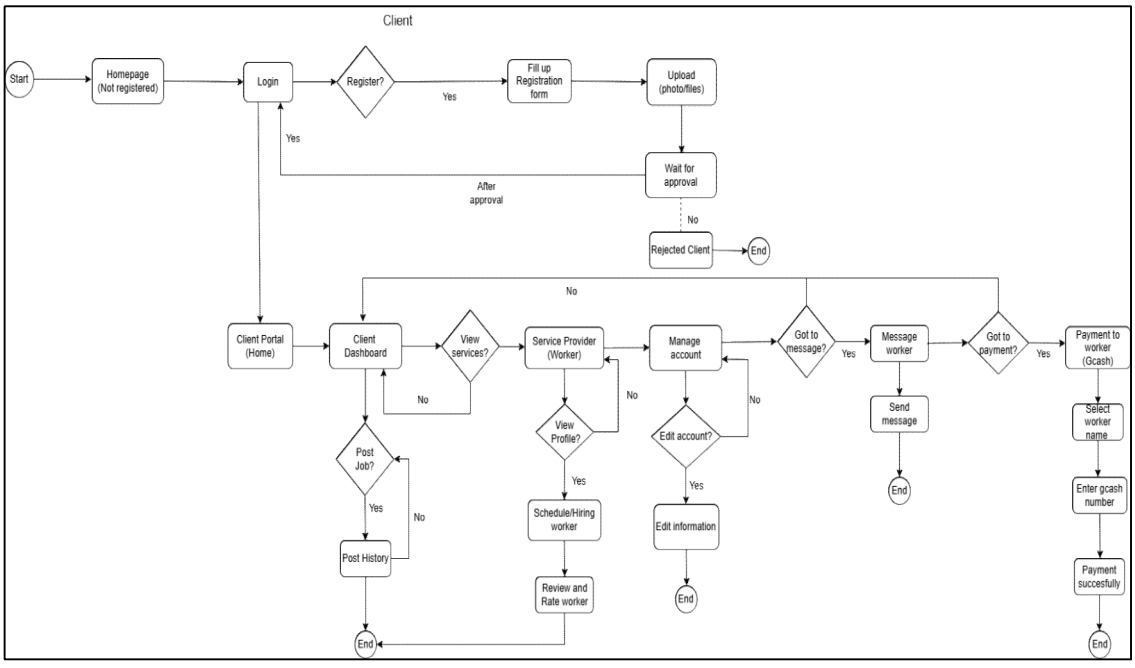
CONCEPTUAL FRAMEWORK



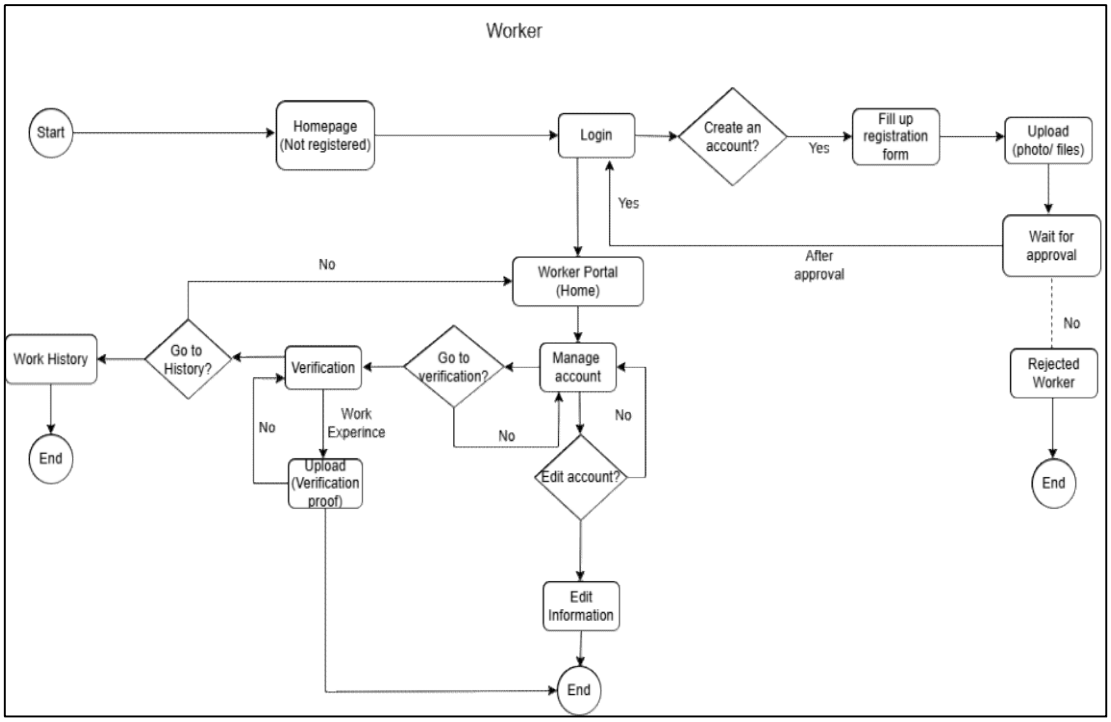
SYSTEM ARCHITECTURE



USE CASE DIAGRAM



SYSTEM FLOWCHART FOR CLIENT



SYSTEM FLOWCHART FOR WORKER

MINIMUM SYSTEM REQUIREMENTS	RECOMMENDED SYSTEM REQUIREMENTS
Processor ryzen 5	Ryzen 4 above 8 gb above
Windows 10	Windows 10
Storage space 16 gb	160Gb

Minimun requirements for laptop/computer

HARDWARE

Minimum requirements for smartphone

APPENDIX B

Source Code

```
head>
ody>
  <div class="login-card">
      <div class="login-header">
          <h2>Client Login</h2>
          <p>Access your account to find skilled workers</p>
      </div>

      <div class="login-body">
          <!-- Display Error Alerts -->
          <?php if (isset($error_message)): ?>
              <div class="alert"><?php echo $error_message; ?></div>
          <?php endif; ?>

          <form action="login_client.php" method="POST">
              <div class="form-group">
                  <label for="username">Username</label>
                  <input type="text" class="form-control" name="username" id="username" placeholder="Enter your username" required>
              </div>

              <div class="form-group">
                  <label for="password">Password</label>
                  <div class="password-input-wrapper">
                      <input type="password" class="form-control" name="password" id="password" placeholder="Enter your password" required>
                      <button type="button" class="toggle-password" onclick="togglePassword()" aria-label="Toggle password visibility">
                          <svg id="eyeIcon" width="20" height="20" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2" stroke-linecap="round" stroke-linejoin="round
                              <path id="eyePath" d="M1 12s4-8 11-8 11 8 11 8-4 8-11 8-11-8-11-8z"></path>
                              <circle id="eyeCircle" cx="12" cy="12" r="3"></circle>
                              <line id="eyeSlash" x1="0" y1="0" x2="0" y2="0" style="display: none;"></line>
                          </svg>
                      </button>
                  </div>
                  <p class="password-note">Your password is case-sensitive</p>
              </div>

              <button type="submit" class="login-btn">Sign In</button>

              <div class="links">
                  <a href="login_worker.php">Login as Worker</a>
                  <a href="forgot_password_client.php">Forgot Password?</a>
              </div>
          </form>

          <div class="register-link">
              Don't have an account? <a href="terms_condition.php">Register Now</a>
          </div>
```

Source Code for login (client)

```
</head>
<body>
    <div class="login-card">
        <div class="login-header">
            <h2>Worker Login</h2>
            <p>Be one of skilled workers</p>
        </div>

        <div class="login-body">
            <!-- Display Error Alerts -->
            <?php if (isset($error_message)): ?>
                <div class="alert"><?php echo $error_message; ?></div>
            <?php endif; ?>

            <form action="login_worker.php" method="POST">
                <div class="form-group">
                    <label for="username">Username</label>
                    <input type="text" class="form-control" name="username" id="username" placeholder="Enter your username" required>
                </div>

                <div class="form-group">
                    <label for="password">Password</label>
                    <div class="password-input-wrapper">
                        <input type="password" class="form-control" name="password" id="password" placeholder="Enter your password" required>
                        <button type="button" class="toggle-password" onclick="togglePassword()" aria-label="Toggle password visibility">
                            <svg id="eyeIcon" width="20" height="20" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2" stroke-linecap="round" stroke-linejoin="rou
                                <path id="eyePath" d="M1 12s4-8 11-8 11 8 11 8-4 8-11 8-11-8-11-8z"></path>
                                <circle id="eyeCircle" cx="12" cy="12" r="3"></circle>
                                <line id="eyeSlash" x1="0" y1="0" x2="0" y2="0" style="display: none;"></line>
                            </svg>
                        </button>
                    </div>
                    <p class="password-note">Your password is case-sensitive</p>
                </div>

                <button type="submit" class="login-btn">Sign In</button>

                <div class="links">
                    <a href="login_client.php">Login as Client</a>
                    <a href="forgot_password_worker.php">Forgot Password?</a>
                </div>
            </form>

            <div class="register-link">
                Don't have an account? <a href="terms_condition_worker.php">Register Now</a>
            </div>
        </div>
```

Source Code for login (worker)

register_client.php

```php
if ($_SERVER["REQUEST_METHOD"] == "POST") {
    $id_picture_extension = strtolower(string: pathinfo(path: $_FILES['id_picture']['name'], flags: PATHINFO_EXTENSION));
    $barangay_clearance_extension = strtolower(string: pathinfo(path: $_FILES["barangay_clearance"]["name"], flags: PATHINFO_EXTENSION));

    $id_picture_name = "id_" . $timestamp . "_" . $random . "." . $id_picture_extension;
    $barangay_clearance_name = "clearance_" . $timestamp . "_" . $random . "." . $barangay_clearance_extension;

    // Validate picture if uploaded via file
    if (isset($_FILES["picture"]) && $_FILES["picture"]["error"] === UPLOAD_ERR_OK) {
        $check_picture = getimagesize(filename: $_FILES["picture"]["tmp_name"]);
        if ($check_picture === false) {
            echo "<script>alert('Picture is not a valid image.'); window.history.back();</script>";
            exit();
        }

        // Check file size
        $maxFileSize = 10 * 1024 * 1024;
        if ($_FILES["picture"]["size"] > $maxFileSize) {
            echo "<script>alert('Picture file is too large. Maximum file size is 10MB.'); window.history.back();</script>";
            exit();
        }

        // Allow only JPG, JPEG, PNG
        $allowed_types = ['jpg', 'jpeg', 'png'];
        if (!in_array(needle: $picture_extension, haystack: $allowed_types)) {
            echo "<script>alert('Only JPG, JPEG, and PNG files are allowed for picture.'); window.history.back();</script>";
            exit();
        }
    }

    // Check file size for other documents
    $maxFileSize = 10 * 1024 * 1024;
    if ($_FILES["id_picture"]["size"] > $maxFileSize || $_FILES["barangay_clearance"]["size"] > $maxFileSize) {
        echo "<script>alert('One of your files is too large. Maximum file size is 10MB per file.'); window.history.back();</script>";
        exit();
    }

    // Allow only JPG, JPEG, PNG for documents
    $allowed_types = ['jpg', 'jpeg', 'png'];
    if (!in_array(needle: $id_picture_extension, haystack: $allowed_types) || !in_array(needle: $barangay_clearance_extension, haystack: $allowed_types)) {
        echo "<script>alert('Only JPG, JPEG, and PNG files are allowed.'); window.history.back();</script>";
        exit();
    }

    // Verify files exist before moving
    if (!isset($_FILES["id_picture"]["tmp_name"]) || !isset($_FILES["barangay_clearance"]["tmp_name"])) {
        echo "<script>alert('Please select all required files.'); window.history.back();</script>";
        exit();
    }
```

Source Code for Register(client)

register_worker.php

```php
function uploadFile($input_name, $allowed_types, $max_size, $required = false): string|null {

        $image_data = $_POST['camera_picture'];
        $image_data = str_replace(search: '', replace: '', subject: $image_data);
        $image_data = str_replace(search: '', replace: '', subject: $image_data);
        $image_data = str_replace(search: ' ', replace: '+', subject: $image_data);
        $decoded_image = base64_decode(string: $image_data);

        file_put_contents(filename: $target_dir . $filename, data: $decoded_image);
        return $filename;
    }

    // Handle regular file upload
    if (empty($_FILES[$input_name]["name"])) {
        if ($required) die("Missing required file: $input_name");
        return null;
    }

    $timestamp = time();
    $random = rand(min: 1000, max: 9999);
    $extension = strtolower(string: pathinfo(path: $_FILES[$input_name]["name"], flags: PATHINFO_EXTENSION));
    $filename = $input_name . "_" . $timestamp . "_" . $random . "." . $extension;
    $target_file = $target_dir . $filename;

    if (!in_array(needle: $extension, haystack: $allowed_types)) {
        die("Invalid file type for $input_name. Allowed types: " . implode(separator: ', ', array: $allowed_types));
    }

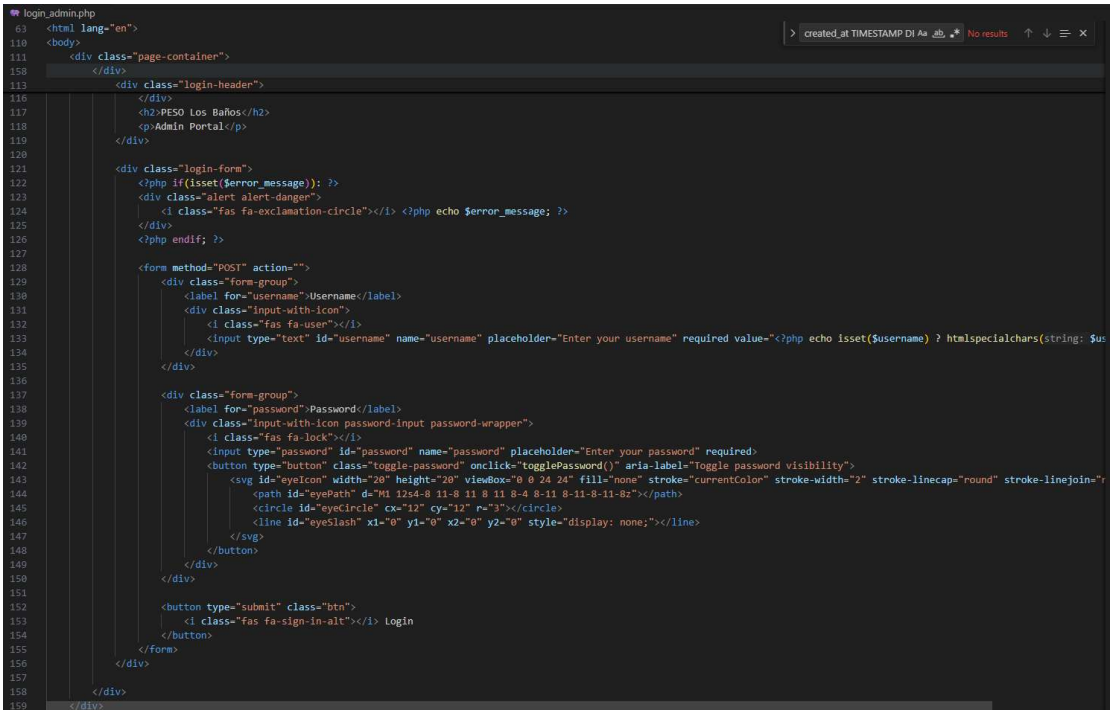
    if ($_FILES[$input_name]["size"] > $max_size) {
        die("File $input_name is too large. Maximum size: " . ($max_size / (1024 * 1024)) . "MB");
    }

    if (!move_uploaded_file(from: $_FILES[$input_name]["tmp_name"], to: $target_file)) {
        die("Failed to upload $input_name.");
    }

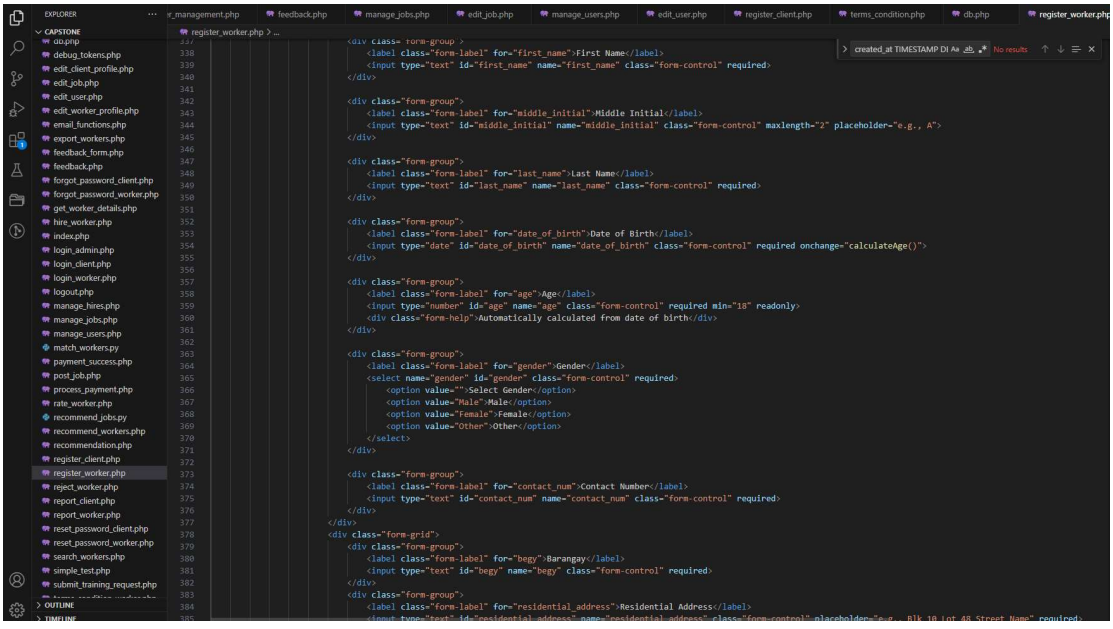
    return $filename;
}

// Upload files with 10MB limit
$picture_name = uploadFile(input_name: "picture", allowed_types: ['jpg', 'jpeg', 'png'], max_size: 10485760, requir…true); // 10MB
$resume_name = uploadFile(input_name: "resume", allowed_types: ['pdf', 'doc', 'docx'], max_size: 10485760); // 10MB
$birth_cert = uploadFile(input_name: "birth_certificate", allowed_types: ['pdf', 'jpg', 'jpeg', 'png'], max_size: 10485760); // 10MB
$nbi_clear = uploadFile(input_name: "nbi_clearance", allowed_types: ['pdf', 'jpg', 'jpeg', 'png'], max_size: 10485760); // 10MB
$brgy_clear = uploadFile(input_name: "barangay_clearance", allowed_types: ['pdf', 'jpg', 'jpeg', 'png'], max_size: 10485760); // 10MB
$voter_id = uploadFile(input_name: "voter_id", allowed_types: ['pdf', 'jpg', 'jpeg', 'png'], max_size: 10485760); // 10MB
$drug_test = uploadFile(input_name: "drug_test_clearance", allowed_types: ['pdf', 'jpg', 'jpeg', 'png'], max_size: 10485760); // 10MB
```

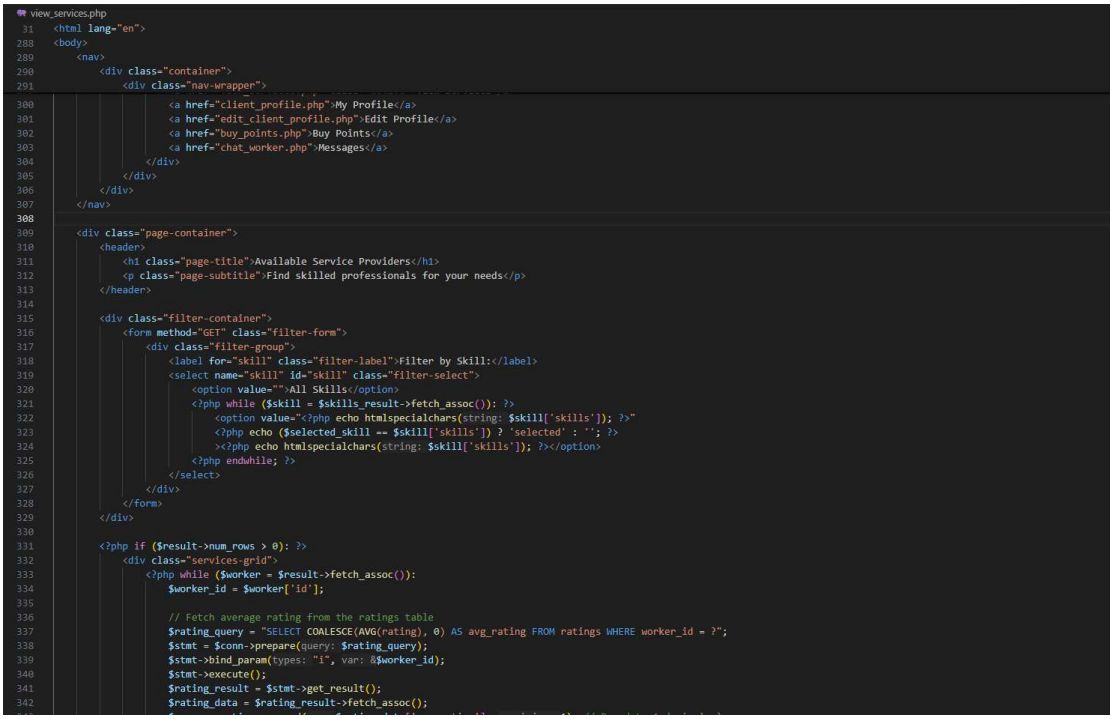
Source Code for Register(worker)



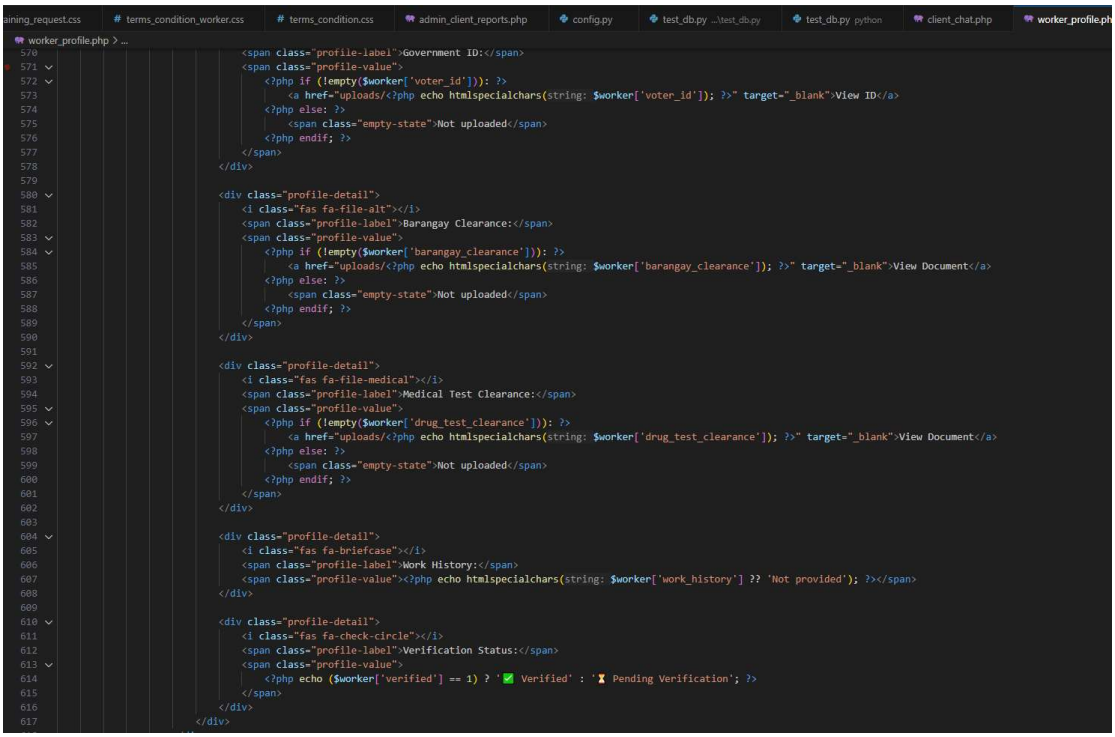
Source Code for Admin Authentication



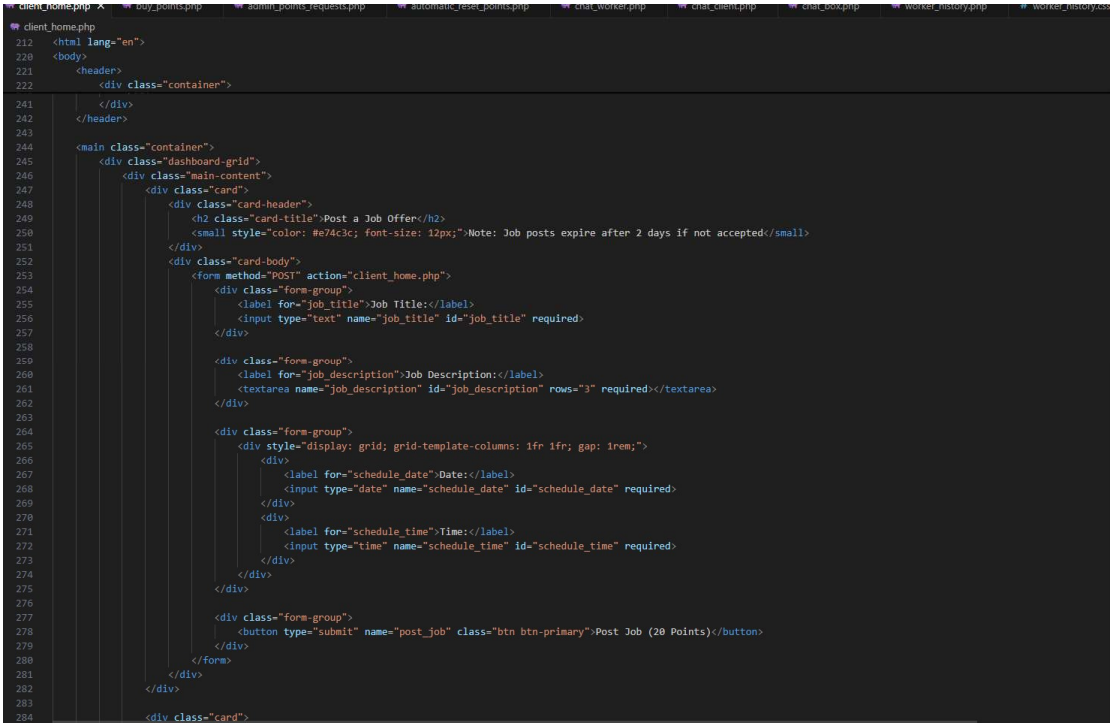
Source Code for Job Application



Source Code for Sorting Job Category



Source Code for Job Seeker Profile Side



```
client_home.php
<html lang="en">
<body>
    <header>
        <div class="container">
        </div>
    </header>

    <main class="container">
        <div class="dashboard-grid">
            <div class="main-content">
                <div class="card">
                    <div class="card-header">
                        <h2 class="card-title">Post a Job Offer</h2>
                        <small style="color: #e74c3c; font-size: 12px;">Note: Job posts expire after 2 days if not accepted</small>
                    </div>
                    <div class="card-body">
                        <form method="POST" action="client_home.php">
                            <div class="form-group">
                                <label for="job_title">Job Title:</label>
                                <input type="text" name="job_title" id="job_title" required>
                            </div>

                            <div class="form-group">
                                <label for="job_description">Job Description:</label>
                                <textarea name="job_description" id="job_description" rows="3" required></textarea>
                            </div>

                            <div class="form-group">
                                <div style="display: grid; grid-template-columns: 1fr 1fr; gap: 1rem;">
                                    <div>
                                        <label for="schedule_date">Date:</label>
                                        <input type="date" name="schedule_date" id="schedule_date" required>
                                    </div>
                                    <div>
                                        <label for="schedule_time">Time:</label>
                                        <input type="time" name="schedule_time" id="schedule_time" required>
                                    </div>
                                </div>
                            </div>

                            <div class="form-group">
                                <button type="submit" name="post_job" class="btn btn-primary">Post Job (20 Points)</button>
                            </div>
                        </form>
                    </div>
                </div>

                <div class="card">
```

Source Code for Job Posting

APPENDIX C

Communication Letters and Approval

APPROVAL SHEET

The Capstone Project entitled **"iServe+: A PESO LOS BAÑOS Service Management Process-Based Hiring Platform with AI-Powered Upskilling Recommendation for Blue-Collar Workers"** prepared and submitted by **Julius M. Donguines, Kristine A. Magpantay, and Abegail Palada** in partial fulfillment of the requirements for the degree Bachelor of Science in Information Technology specialized in Service Management Program, is hereby recommended for approval and acceptance.

CRISANTO F. GULAY
Adviser

Approved and Accepted by the Committee on Oral Examination with a grade of_____.

JONALYN JOY B. LABAYNE
Member

SHERWIN B. SAPIN
Member

JOCELYN O. PADALLAN
Member

JEFFERSON L. LERIOS
Member

Accepted in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology specialized in Service Management Program.

LOYD S. ECHALAR
Associate Dean, College of Computer Studies

OFELIA B. MANINGAS, EdD.
Chairperson, Research and Development Office

_____
Date Signed

RESEARCH CONTRIBUTION NO.	

CERTIFICATION OF APPROVAL

This is to certify that after having satisfactorily completed all the requirements of the capstone project entitled:

“iServe+: A PESO LOS BAÑOS Service Management Process-Based Hiring Platform with AI-Powered Upskilling Recommendation For Blue-collar Worker”

JULIUS M. DONGUINES, KRISTINE A. MAGPANTAY & ABEGAIL PALADA has been granted permission to proceed with their final presentation.

As their Research Adviser, I confirm that the proponents have met the academic standards and requirements set forth by the department, and therefore, I approve that their request to conduct the final presentation.

Mr. Crisanto F. Gulay

Capstone Project Adviser

2026

2026

HON. MAYOR NEIL ANDREW N. NOCON

Municipal Mayor

Los Baños, Laguna

Thru: Gliceria A. Trinidad

Senior Labor and Employment Officers III

Subject: **Endorsement of Capstone Project - A Mobile Application for PESO**

Dear Hon. Nocon and Ms. Trinidad,

I am writing to you as the adviser of **Mr. Julius M. Donguines, Ms. Kristine A. Magpantay and Ms. Abegail Palada**, a BSIT **Student of Laguna State Polytechnic University of Los Baños** who have been diligently working on their capstone project entitled **"iServe+: A PESO LOS BAÑOS Service Management Process-Based Hiring Platform with AI-Powered Upskilling Recommendation for Blue-Collar Workers".**

This project aims to streamline the operation of **PESO** and a more efficient and user-friendly platform for the public. The application is designed with deep understanding of the needs of **PESO** and the people it serves.

Mr. Donguines, Ms. Magpantay, and Ms. Palada have shown exceptional dedication and commitment to this project. Their innovative approach and technical skills have resulted in a mobile application that is believed to greatly benefit PESO and the community.

This project is wholeheartedly endorsed and recommended for due consideration for implementation. There is strong confidence that this application significantly contributes to the efficiency and effectiveness of PESO services.

Thank you for your attention to that matter. We look forward to the opportunity to present this project t you and discuss how it can benefit **PESO** and the community of **Los Baños, Laguna**.

Sincerely,

Mr. Crisanto F. Gulay

Capstone Project Adviser

APPENDIX D

Publishable Paper

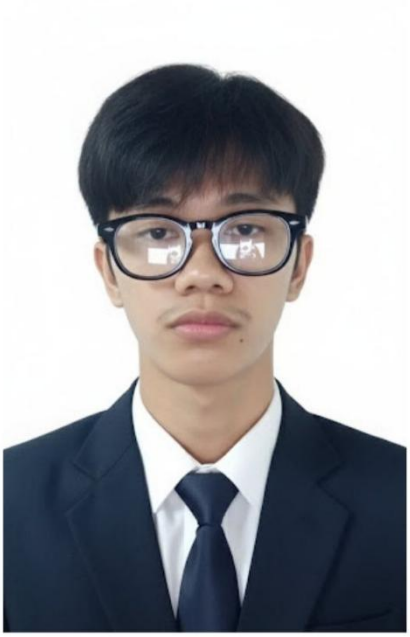
APPENDIX E

Plagiarism Results

APPENDIX F

Curriculum Vitae

Curriculum Vitae



JULIUS DONGUINES

Student

09166923071

juliusdonguines2018@gmail.com

Brgy. Bañadero, Calamba Laguna

PROFILE

Web Developer and IT student majoring in Service Management, with experience in website development, database management, and system design. Skilled in building functional web platforms, designing user-friendly interfaces, and managing data-driven systems. Developed a PESO Los Baños program that connects blue-collar workers with clients for part-time job opportunities. Highly motivated to apply technical knowledge and problem-solving skills to real-world IT projects and community-focused solutions.

EDUCATION

2022-Present

Bachelor of Science,
Information Technology
Major in Service Management Program
Laguna State Polytechnic University

SKILLS

- Web Development (HTML, CSS, JS, PHP)
- Database Management (MySQL)
- UI/UX Design
- System Development
- API Integration
- Troubleshooting & Backend Logic



Curriculum Vitae

KRISTINE MAGPANTAY

STUDENT

09649956549

iamkristine233@gmail.com

Tuntungin Putho, Los Baños, Laguna

PROFILE

I am a Programmer and Designer seeking opportunities in that field, where I can apply my skills for continuous improvement

EDUCATION

2022-Present

Bachelor of Science,
Information Technology
Major in Service Management Program

Laguna State Polytechnic University

SKILLS

- Programmer
- Designer

Curriculum Vitae



ABEGAIL PALADA

STUDENT

0985-156-4914

paladaabegail32@gmail.com

Brgy. Real Calamba City Laguna

PROFILE

An aspiring Front-End Developer and IT student with experience in service management, customer support, and technical operations. Skilled in building responsive user interfaces, providing system assistance, and collaborating with teams to improve workflow efficiency. Highly motivated to apply technical knowledge and problem-solving skills in real-world IT environments.

EDUCATION

2022-Present

Bachelor of Science,
Major in Service Management Program

Laguna State Polytechnic University

SKILLS

- HTML, CSS, JavaScript
- Customer Service & Communication
- Documentation & Reporting
- Problem-Solving
- Designer / Basic Graphic Design

WORK EXPERIENCES

2021-2022

Acbel Polytech Inc,.

Production Operator

- Collaborated with team members to ensure smooth workflow and solve minor operational issues.

2023-2023

Puregold

Cashier

- Handled POS operations and managed daily cash transactions.
- Provided customer service and resolved purchase-related concerns.
- Ensured accurate recording, reporting, and balancing of sales.
- Maintained efficiency and professionalism during peak hours.